

**FINGERPRINT BIOMETRIC ATTENDANCE SYSTEM: MOUNTAIN TOP
UNIVERSITY AS CASE STUDY**

BY

**NDUKWE BENEDICT CHIBUIKEM
GWU/BSC/CS/2020/36182**

A PROJECT

**SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENTS
FOR THE AWARD OF
BACHELOR OF SCIENCE DEGREE**

**DEPARTMENT OF COMPUTER SCIENCE AND SOFTWARE
ENGINEERING**

(GLOBAL WEALTH UNIVERSITY)

SUBMITTED TO:

GLOBAL WEALTH UNIVERSITY, LOME, TOGO

AUGUST, 2026

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CERTIFICATION

The undersigned have examined the project entitled 'FINGERPRINT BIOMETRIC ATTENDANCE SYSTEM: MOUNTAIN TOP UNIVERSITY AS CASE STUDY', presented by 'NDUKWE BENEDICT CHIBUIKEM', a candidate for the degree of Bachelor of Science (Computer Science and Software Engineering) and hereby certify that it is worthy of acceptance.

Date
Signature

.....

Engr. Ebofuai Joshua

Supervisor's

Date

.....

Coordinator's Signature

Date

.....

External Examiner's Signature

DECLARATION

I 'NDUKWE BENEDICT CHIBUIKEM', hereby declare that this project is original and has been written by me and has not been submitted to any institution.

.....

Date & Signature

DEDICATION

This project is dedicated to God Almighty, the source of all wisdom and strength, who made this academic journey possible. It is also dedicated to my family, whose unwavering prayers, encouragement, and support sustained me through every stage of this work.

ACKNOWLEDGEMENTS

Great thanks to God Almighty for His grace and guidance throughout this academic programmer.

I am sincerely grateful to my project supervisor, Engr. Joshua Ebofuai, for his patience, guidance, and constructive feedback throughout the course of this research. His supervision shaped this work considerably.

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To my family, I owe a great debt of gratitude for your prayers, encouragement, and support throughout my time at university. To my friends and colleagues who offered assistance in any form during the course of this work, I am sincerely grateful.

ABSTRACT

This study designed a fingerprint biometric attendance system for Mountain Top University, Ogun State, Nigeria. The problem motivating the study is the widespread use of manual attendance recording in Nigerian universities, a process that is vulnerable to impersonation, consumes class time, produces unreliable records, and places a heavy administrative burden on staff at the end of each semester. These weaknesses are particularly significant because the National Universities Commission requires students to meet a minimum attendance level of 75 percent before they can be certified eligible for examinations. The study adopted a design and development research approach, relying entirely on secondary data drawn from twenty-four peer-reviewed studies on fingerprint and other automated attendance systems, each verified individually before use. The theoretical framework is built on minutiae-based fingerprint matching theory, and the system design follows the Object-Oriented Analysis and Design (OOAD) methodology, with diagrams produced using the Unified Modeling Language (UML). The proposed system consists of four components: an enrollment module that captures and stores fingerprint templates alongside user details, a verification module that checks each scan against stored records before marking attendance, a database that organizes all user, course, and attendance data, and a reporting module that generates attendance summaries and calculates each student's attendance percentage against the 75 percent rule. The design includes a flagging mechanism for cases where a fingerprint cannot be matched, addressing a gap found in the reviewed literature. The study is limited to the design stage and does not build or test a working system. It concludes that a fingerprint biometric attendance system is a well-supported and practically achievable response to the weaknesses of manual attendance recording at Mountain Top University. Keywords: Fingerprint recognition, biometric attendance system, Mountain Top University, minutiae-based matching, OOAD, UML, design-only study, National Universities Commission.

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LIST OF ABBREVIATIONS

EER	- Equal Error Rate
ERD	- Entity Relationship Diagram
ESP32	- Espressif Systems ESP32 Microcontroller
FAR	- False Acceptance Rate
FRR	- False Rejection Rate
GSM	- Global System for Mobile Communications
IoT	- Internet of Things
MTU	- Mountain Top University
NUC	- National Universities Commission
OOAD	- Object-Oriented Analysis and Design
RFID	- Radio Frequency Identification
SMTP	- Simple Mail Transfer Protocol
UML	- Unified Modeling Language

CHAPTER ONE

INTRODUCTION

1.1 Background to the Study

Attendance has always carried more weight in academic life than the simple act of writing a name on a sheet of paper suggests. It is used to judge engagement, to enforce discipline, and, in most Nigerian universities, to decide whether a student is even permitted to sit for an examination. The National Universities Commission requires students to maintain a minimum attendance level, commonly set at 75 percent of classes held, before they can be certified eligible for examinations (Mikailu, Faseke, Ishaya and Ahmed, 2025). That single requirement turns attendance from a background administrative task into something that directly shapes a student's academic future, which is exactly why the method used to record it matters as much as it does.

For most of the history of formal education, that method has been the paper register. A lecturer calls names, or passes a sheet around the room, and students confirm their presence with a signature. This approach has an obvious appeal: it needs no equipment, no electricity, and no training to use. Its weaknesses, though, become more visible as class sizes grow. A register that works fine for a class of thirty becomes unwieldy for two hundred, and the paper itself, being paper, can be misplaced, torn, or left behind in a lecture hall by a distracted lecturer moving on to the next class.

The paper register was not the last word on attendance recording, even before biometrics arrived. Some institutions moved to barcode cards, where each student carries a card scanned at the start of class, or to magnetic stripe cards read the same way a bank card is read. Both were an improvement in speed over calling names one by one, but neither solved the underlying problem this study is concerned with, a card, unlike a fingerprint, can be lent to a friend, which means the risk of one person marking attendance for another does not go away, it simply moves from a signature to a piece of plastic.

The difficulty is not unique to Nigeria. Althothaily, Alradaey, Oqbah and El-Kustaban (2015) documented the same pressure at a university in Yemen, where manual roll calls consumed

lecture time and made it nearly impossible to track whether students had met the minimum attendance threshold for examination eligibility. Rahman, Rahman and Rahman (2018) described an identical situation at a university in Bangladesh, where the same combination of time loss, inaccurate records, and impersonation drove the development of a fingerprint-based alternative. That universities on three different continents, operating in very different institutional environments, keep arriving at the same diagnosis of the same manual-system problem is worth stating plainly, because it means the case for an automated alternative does not rest on any one country's experience alone.

Nigerian universities, private and public alike, have not been shielded from these pressures. Enrollment has grown faster than administrative systems in many institutions, and the paper register, largely unchanged since it was first adopted, has been asked to do more with less support than ever. Mountain Top University, a private university located along the Lagos-Ibadan Expressway in Ogun State and accredited by the National Universities Commission, sits within this same context. Like most Nigerian universities, it depends on some form of attendance monitoring to enforce examination eligibility, and there is no particular reason to think the pressures documented elsewhere in Nigerian higher education stop at its gates.

Private universities occupy a particular place in this picture. Since the federal government opened university education to private proprietors in 1999, the number of private universities in Nigeria has grown quickly, and a good number of them, Mountain Top University included, are still relatively young institutions, building out their administrative systems at the same time as they grow their student population. An older public university may have decades of accumulated staffing and procedure to fall back on when a manual process starts to strain under rising numbers. A newer private university does not necessarily have that cushion, which means the same pressures documented above, a paper-based process built for a smaller cohort being asked to handle a growing one, can show up earlier in a young institution's life than they would somewhere with a longer administrative history.

The growth of private universities in Nigeria since 1999 has also meant that a large number of them are operating simultaneously in a phase where their student populations are still expanding and their administrative infrastructure is still being built out. This combination, growing student numbers and developing administrative systems, is precisely the environment in which manual

attendance recording becomes most difficult to sustain without problems. A system that might work adequately when a university has five hundred students in its first few years can become genuinely unreliable by the time that same institution reaches two or three thousand, yet the paper register rarely changes to reflect this shift.

Biometric technology has emerged as one of the more credible responses to these pressures, and fingerprint recognition in particular has become the most widely adopted form of it in attendance systems. The appeal is straightforward. A fingerprint is fixed at birth and does not change meaningfully over the course of a person's life, which means it can serve as a form of identification that a signature, a spoken name, or a piece of paper cannot easily imitate. Where a signature can be forged in seconds and a name can be called out by anyone sitting near an absent classmate, a fingerprint belongs to exactly one person and no one else.

This is not a theoretical claim. Elebede, Olusanya, Sodipo and Sam-Leeloo (2024) designed and built a fingerprint-based student attendance system and tested it directly against the paper method it was meant to replace, reporting that it recorded attendance with far greater consistency and left little room for the kind of manipulation that paper registers allow. Their work forms part of a wider pattern across Nigerian institutions experimenting with the same basic idea, that a biometric identifier can do a more reliable job than a written signature.

Akinduyite and colleagues (2013) reported much the same thing over a decade earlier, testing a fingerprint system on more than a hundred real staff and students at the Federal University of Technology, Akure, and finding that it prevented impersonation while working faster than the manual method it replaced. Odia and colleagues (2020) reached a similar conclusion more recently at the University of Benin, adding a feature not seen in the earlier work, their system could automatically flag students who had fallen short of the attendance percentage needed for examination eligibility, without a staff member having to work this out by hand. That researchers a decade apart, at different institutions, kept arriving at the same basic result is itself worth noting.

Olasupo and colleagues (2022) tested this idea at Olabisi Onabanjo University, building a fingerprint attendance system and putting it to use in real classes over a semester. Lecturers who used the system reported that it was accurate, fast, and dependable, while students said it made

attendance fraud, one person signing in for another, effectively impossible. What stands out about this study is that it was not run as a controlled experiment removed from ordinary university life; it was deployed in real classrooms with real students and lecturers, and its conclusions came from people who actually depended on the outcome.

Fingerprint recognition is not the only biometric route universities have tried. Olanipekun and Boyinbode (2015) took an RFID-based approach instead, giving each student a tag that a reader could detect automatically as they entered a classroom. Their system worked, and it solved several of the same problems that fingerprint systems address, but it introduced a different vulnerability: a tag can be handed to someone else in a way that a fingerprint cannot. This difference matters for understanding why fingerprint recognition has remained the more common choice for attendance specifically, as opposed to applications like inventory tracking, where RFID is harder to beat.

More recently, some Nigerian researchers have turned to facial recognition instead. Mikailu, Faseke, Ishaya and Ahmed (2025) built a system around this technology for Nigerian university classrooms and reported strong accuracy figures for both detecting and correctly identifying students. Their stated motivation for building the system in the first place is worth pausing on: they pointed to proxy attendance and the administrative burden of manual methods as the exact problems driving universities toward automation. That a 2025 study is still citing the same underlying problems that RFID researchers described a decade earlier, and that fingerprint researchers described in between, suggests these are not isolated complaints but a persistent feature of manual attendance systems generally.

Fingerprint recognition also compares favorably to other biometric modalities on practical grounds. Iris recognition is highly accurate, but the scanning equipment it requires is expensive and less widely available in Nigeria than fingerprint sensors. Voice recognition is vulnerable to background noise in a busy lecture hall. Facial recognition, as Mikailu, Faseke, Ishaya and Ahmed (2025) demonstrated, can work well, but it depends on consistent lighting and camera positioning across every lecture room a university operates. Fingerprint sensors, by contrast, are inexpensive and largely unaffected by lighting or ambient noise, which is part of why they have remained a practical default even as other biometric options have matured.

Taken together, this body of work establishes two things clearly. First, that manual attendance recording carries specific, repeatedly documented weaknesses that show up across different Nigerian universities and different points in time. Second, that biometric systems, fingerprint recognition among them, have been built and tested as workable responses to those weaknesses, not as speculative or untested ideas. This study takes Mountain Top University as its point of reference and proposes a fingerprint biometric attendance system designed around the specific problems that this existing body of research has already identified, rather than around assumptions about what might be wrong with the current process.

1.2 Profile of Mountain Top University

Mountain Top University was established in 2015 by the Mountain of Fire and Miracles Ministries, a Pentecostal Christian denomination founded by Dr Daniel Kolawole Olukoya, who also serves as the university's chancellor (Mountain Top University, n.d.). The university sits at Km 12 on the Lagos-Ibadan Expressway, in Prayer City, Ogun State, on land adjoining the ministry's own campus. It holds full operating status from the National Universities Commission, and, like every Nigerian university under the commission's oversight, its academic programmes are periodically accredited against national minimum standards, the same regulatory relationship, and the same attendance-linked examination eligibility rule, described in the section above.

The university runs undergraduate programmes across several colleges spanning the basic and applied sciences, the humanities, management, and the social sciences, with a smaller postgraduate school layered on top. Each college runs its own set of departments, and each department manages its own lecture timetable, which means attendance is not a centrally coordinated activity at Mountain Top University any more than it is at most Nigerian universities, it is managed separately by each lecturer across each course they teach. That decentralised reality is part of what makes a manual process difficult to supervise or standardise at the institutional level.

Its undergraduate population sits in the region of a couple of thousand students, a figure well below the size of Nigeria's largest public universities but still large enough that a manual, paper-based attendance process is not a minor administrative detail, particularly in departments running sizeable first- and second-year classes. A university of this size will typically have lecture groups

ranging from small seminars of twenty or thirty students to larger survey courses with over a hundred, and the paper register performs differently across these settings. A small class may produce a manageable register that the lecturer can keep track of without difficulty. A large one puts the same process under a pressure the register was not designed for.

Mountain Top University's stated mission ties academic training to moral and spiritual formation as much as to technical competence, and its administrative culture, attendance recording included, sits inside that same institutional emphasis on discipline and conduct (Mountain Top University, n.d.). This does not exempt the university's own attendance process from the weaknesses documented throughout this chapter. Nothing about the institution's public profile suggests its paper-based process would somehow avoid the impersonation, lost time, and record-keeping problems reported at other Nigerian universities of comparable size. If anything, it means an accurate attendance record carries slightly more institutional weight at Mountain Top University than it might elsewhere, since attendance and conduct are treated there as connected concerns rather than separate ones.

Attendance data at Mountain Top University, as at most Nigerian universities, does not stay with the lecturer who collected it. It moves outward, from the lecturer's own record to the departmental office, from the departmental office to the faculty registry, and from there to whatever central system decides examination eligibility at the end of the semester. Each handover in that chain is a point where a handwritten record can be misread, miscopied, or simply lost. A digital system that captures a record at the moment of verification and stores it in one place accessible to all these levels at once removes the handover risk entirely, which is a practical advantage that goes beyond simply being faster than passing a sheet around a room.

1.3 Statement of the Problem

The problems associated with manual attendance recording are not vague or hard to pin down. Olanipekun and Boyinbode (2015) documented them directly: attendance records become difficult to preserve intact over the course of a semester, class time is lost to the physical act of passing sheets around and waiting for signatures, documentation ends up incomplete or inconsistent from one class to the next, students forget to sign in even when present, lecturers

occasionally leave the attendance sheet behind in a previous lecture hall, and, perhaps most seriously, students sign in on behalf of classmates who are not actually present.

That last problem deserves particular attention, because it is not simply an inconvenience, it is a failure of the entire system's purpose. If attendance can be recorded for a student who never walked into the room, then the record no longer measures presence at all. It measures only how willing someone nearby was to write an extra name. Mikailu, Faseke, Ishaya and Ahmed (2025) identified this exact issue, proxy attendance, as one of the central weaknesses driving the search for automated alternatives in Nigerian universities, and their framing of the problem lines up closely with what Olanipekun and Boyinbode described a decade before them.

The consequences of this weakness extend beyond the classroom itself. Because Nigerian universities generally tie examination eligibility to a minimum attendance threshold, an inaccurate attendance record does not just misrepresent who was present on a given day, it can misrepresent whether a student has actually met the requirement to sit for an exam at all. A student who benefited from proxy attendance may be certified eligible despite having missed classes they were supposed to attend, while the administrative staff responsible for compiling these records have no reliable way of catching the discrepancy after the fact, since the paper trail looks complete even when it is not accurate.

Manual recording also places a burden on the people responsible for maintaining it, a burden that grows heavier as student numbers increase. Compiling a semester's worth of handwritten attendance data into a usable summary takes time, and every step of that process, transcription, addition, cross-referencing against course registration lists, introduces a fresh opportunity for error. Once a mistake has been made, it is often difficult to trace back to its source, since the original paper records rarely include enough detail to reconstruct exactly where something went wrong. Olasupo and colleagues (2022) found that lecturers who switched to a fingerprint-based system described it as considerably less burdensome than the paper process it replaced, a comment that reflects as much on the demands of the old system as it does on the convenience of the new one.

There is no indication that Mountain Top University is exempt from these pressures. It is a Nigerian university operating under the same National Universities Commission attendance

requirements described above, and it depends, like most of its peers, on some version of the same manual recording process that researchers elsewhere have already found wanting. The problem this study addresses, then, is not unique to Mountain Top University in the sense of being locally invented, but it is a problem that Mountain Top University shares with the wider set of institutions this literature describes, and one that a fingerprint biometric attendance system has repeatedly been shown, elsewhere, to help resolve.

There is a further consequence worth setting out on its own, since it sits above the level of any single student's record. National Universities Commission accreditation exercises look, among other things, at how consistently an institution can demonstrate that its examination eligibility decisions are properly grounded, and an attendance system that produces records an institution cannot fully trust does not just create disputes at the level of an individual student, it weakens the institution's own position when it has to account for how those decisions were reached. A private university still building its institutional reputation, as Mountain Top University is, has less room to absorb that kind of weakness than an older, more established institution might.

This study responds to that shared problem by proposing a fingerprint biometric attendance system for Mountain Top University, designed specifically around the weaknesses that manual recording has already been documented to have, rather than around assumptions about what a hypothetical future implementation might need.

It is worth being clear that the problem this study is responding to has two sides, not one. The more visible side is impersonation, one student signing in for another. The less visible but equally real side is the student who was genuinely present but whose record ends up wrong anyway, because the attendance sheet was misread when it was transcribed, because an arithmetic error crept into the semester total, or because a sheet was lost between the lecture hall and the departmental office. A fingerprint system addresses both sides at once, since it produces a record that is both harder to falsify and harder to accidentally corrupt than a handwritten one.

1.4 Aim and Objectives of the Study

The aim of this study is to design a fingerprint biometric attendance system for Mountain Top University that addresses the recurring weaknesses associated with manual attendance recording. To achieve this aim, the study sets out to:

- i. examine existing literature on fingerprint biometric attendance systems in order to identify the problems these systems have been built to solve and the approaches previous researchers have taken to solve them;
- ii. analyze the manual attendance recording process typically used in Nigerian university settings, along with the specific weaknesses documented in connection with it;
- iii. design the architecture of a fingerprint biometric attendance system suited to a university environment, including its database structure, its process flow from enrollment through verification, and the relationships between its component modules;
- iv. produce supporting system design documentation, including use case diagrams and entity relationship diagrams, capable of guiding a future implementation of the proposed system at Mountain Top University or a comparable institution.

1.5 Research Questions

This study is guided by the following questions:

- i. What problems are commonly associated with manual attendance recording in university settings, according to existing literature?
- ii. How have fingerprint biometric attendance systems been designed and applied in comparable university settings to address these problems?
- iii. What system architecture and design would be appropriate for a fingerprint biometric attendance system proposed for Mountain Top University?

Each of these questions is answered through the methodology this study adopts. The first is answered in the literature review, the second through a comparative reading of related systems already implemented elsewhere, and the third through the system design work carried out in a later chapter of this project.

1.6 Significance of the Study

This study is relevant to several groups with a direct or indirect stake in how attendance is recorded at Mountain Top University. For students, a well-designed fingerprint biometric system would remove the possibility of another student signing in on their behalf, which protects the

meaning of their own attendance record and removes any ambiguity about their standing with respect to examination eligibility. A student who genuinely attends every class would no longer have to worry that a poorly kept paper record, rather than their own conduct, is what determines whether they are certified eligible for an examination.

For lecturers, such a system would reduce the time spent on the physical mechanics of collecting attendance during class and the administrative work of compiling records afterward, freeing that time for teaching-related tasks instead. A lecturer would also no longer need to personally recall which student's handwriting matched which name on a quiz script, the kind of informal detective work Chapter Two shows some lecturers have resorted to under the current manual process.

For the university's administration, an accurate and consistent attendance record supports better-informed decisions about examination eligibility and reduces the disputes that arise when a student's attendance status is called into question at the end of a semester. Where a paper-based dispute often comes down to one person's word against another's, a fingerprint-verified record gives the administration a firmer basis for resolving the matter one way or the other. Beyond individual disputes, a system that produces verifiable, timestamped attendance records strengthens the institution's overall position when it is required to demonstrate to external bodies, including the National Universities Commission, that its examination eligibility decisions are properly supported by documentation.

More broadly, this study contributes to the growing body of research on biometric attendance systems in Nigerian universities by proposing a design grounded in the documented needs of a specific institution rather than a generic or hypothetical one, which may make its findings more directly useful to Mountain Top University than a more general study would be. Finally, this study may serve as a reference point for future students undertaking related project work, particularly those working within similar constraints of a design-only, literature-based methodology.

To the researcher, this study offers a chance to work through, in a practical setting, exactly the kind of problem system design work is meant to solve, taking a real institutional weakness, checking what has already been tried elsewhere, and turning that into a specific, defensible design rather than a general discussion of what a fingerprint system could theoretically do. To

the wider body of knowledge on attendance systems in Nigerian higher education, this study adds one more data point to an already consistent pattern, that manual recording carries the same weaknesses across different institutions and different points in time, while also documenting openly what a literature-only design process looks like when direct institutional access is not available, a situation this study is unlikely to be the last to encounter.

That last point is worth expanding briefly. Most of the literature this study draws on was produced by researchers who had direct access to the institutions they studied. This study does not have that access, and it says so plainly in Chapter Three rather than working around it. The value of being explicit about this is that it makes the study's reasoning easier to follow and easier to challenge, which is what a well-documented piece of academic work is supposed to allow. A future researcher who finds a flaw in the reasoning, or who has direct access to Mountain Top University, can build on this work in a way they could not if the methodology had been left vague.

1.7 Scope of the Study

This study is limited to the design of a fingerprint biometric attendance system proposed for Mountain Top University. The work covers a review of existing literature on fingerprint biometric attendance systems, an analysis of the weaknesses associated with manual attendance recording, and the production of a system design, including its architecture, database structure, and supporting diagrams such as use case diagrams and entity relationship diagrams.

The study does not extend to building, coding, or physically testing a working system. It does not involve fieldwork, interviews, or questionnaires administered to staff or students at Mountain Top University, since direct access to the institution for this purpose was not available at the time of the study. Mountain Top University is used here as a reference case to ground the proposed design in a specific institutional context, informed by its known status as a Nigerian university subject to National Universities Commission attendance requirements, rather than as a site of direct data collection.

Being specific about what design-only means here is useful, since the term can cover a range of different outputs. In this study, the design produced in Chapter Four consists of a system architecture description, a use case diagram and written explanation of each use case, an entity

relationship diagram, a database design showing the fields and data types for each table, and a process flow covering both enrollment and attendance verification. These are the kinds of outputs that would normally precede the coding stage of a software development project. They are sufficient to communicate how the proposed system would work and how its data would be organised, but they do not include working code, a prototype, screen mockups, or any output that requires the system to have been built or tested.

1.8 Limitations of the Study

The most significant limitation of this study is the absence of primary data collected directly from Mountain Top University. Without direct access to the institution, this study cannot report on the specific attendance practices, class sizes, or administrative arrangements unique to Mountain Top University beyond what is publicly known about the institution generally. The proposed system design is therefore based on documented problems and solutions from comparable Nigerian university settings, rather than on problems verified to exist in this exact form at Mountain Top University itself.

A related limitation concerns the scope of implementation. Because this study is design-only, it cannot report on how the proposed system would actually perform once built, including matters such as real-world fingerprint matching accuracy under local conditions, hardware reliability, or user acceptance among Mountain Top University's own students and staff. These questions would need to be addressed by a future study with the access and resources to carry out an actual implementation and test it under real conditions.

A further limitation follows from relying on literature drawn from other Nigerian universities to inform a design proposed for Mountain Top University specifically. Each university differs in class size, existing infrastructure, and administrative practice, and a design informed by Olabisi Onabanjo University's experience, for instance, will not necessarily transfer without adjustment. This study treats the reviewed literature as a well-evidenced starting point for design, not as a substitute for institution-specific data a future study with direct access would be better positioned to collect.

A final limitation relates to the currency of the information used about Mountain Top University itself. The institutional details presented in section 1.2 above, the university's founding year, its

colleges, its mission, and its approximate student population, were drawn from publicly available information at the time of writing. Since this study had no direct contact with the institution, it cannot confirm whether any of these details have changed since they were recorded, or whether the university has already begun addressing its attendance recording process in ways that are not reflected in the public record. A future study with direct institutional access would be better placed to verify the current state of affairs before applying this study's design to it.

1.9 Definition of Terms

Attendance Management System: a system, manual or automated, used to record and track whether individuals are present at a scheduled event or activity, such as a lecture.

Biometrics: the measurement and use of a person's unique physical or behavioral characteristics, such as a fingerprint or facial structure, for identification or verification purposes.

Enrollment: the process by which a new user's fingerprint is captured and stored in the system for the first time, creating the reference template against which future scans will be compared.

Fingerprint Recognition: a biometric method that identifies or verifies an individual based on the unique pattern of ridges and valleys present on their fingertip.

False Acceptance Rate (FAR): the proportion of verification attempts in which the system incorrectly accepts a fingerprint that does not actually match the claimed identity. A lower false acceptance rate means the system is less likely to let the wrong person through.

False Rejection Rate (FRR): the proportion of verification attempts in which the system incorrectly rejects a fingerprint that does genuinely match the claimed identity, usually because of a poor-quality scan. A lower false rejection rate means fewer legitimate users are turned away.

Minutiae: the specific points on a fingerprint, such as ridge endings and bifurcations, that fingerprint recognition systems analyze and compare in order to match one fingerprint against another.

Authentication: the process of confirming that an individual is who they claim to be, typically by comparing a presented credential, such as a fingerprint, against a stored record.

Verification: a one-to-one comparison process that checks whether a presented fingerprint matches a specific, already claimed identity.

Identification: a one-to-many comparison process that checks a presented fingerprint against an entire database of stored fingerprints in order to determine whose it is, without a prior claim of identity.

Proxy Attendance: a situation in which one individual records attendance on behalf of another individual who is not actually present.

Entity Relationship Diagram (ERD): a diagram used in system design to represent the data entities within a system and the relationships between them.

Use Case Diagram: a diagram used in system design to represent the interactions between users and a system, showing the specific actions a user can perform.

National Universities Commission (NUC): the federal government body responsible for the accreditation and regulation of university education in Nigeria, including the enforcement of minimum attendance requirements for examination eligibility.

1.10 Organization of the Study

This study is organized into five chapters. Chapter One has introduced the background to the study, presented a profile of Mountain Top University, stated the problem the study addresses, and outlined its aim, objectives, research questions, significance, scope, and limitations. Chapter Two reviews existing literature on fingerprint biometric attendance systems and related technologies, alongside the theoretical concepts underlying fingerprint recognition. Chapter Three describes the methodology adopted for this study, including the reasoning behind a secondary, design-based approach in the absence of direct access to Mountain Top University. Chapter Four presents the proposed system design, covering its architecture, database structure, and supporting diagrams. Chapter Five summarizes the study, draws conclusions from the design work carried out, and offers recommendations for any future implementation of the proposed system.

CHAPTER TWO

LITERATURE REVIEW

2.1 Introduction

This chapter reviews work that is relevant to the design of a fingerprint biometric attendance system for Mountain Top University. It starts with a conceptual framework that explains the main ideas the study is built on, followed by a theoretical framework that explains how fingerprint matching actually works and how well it works. After that, the chapter reviews twenty-four studies that were checked one by one to confirm they are real and distinct, grouped first around fingerprint systems built for Nigerian universities, then fingerprint systems built outside Nigeria, and finally other approaches, RFID and facial recognition, that researchers have tried instead. A critical look at this literature follows, pointing out not just what these studies found but how strong that evidence really is. The chapter then states the specific gap this study is filling, why this review matters to the rest of the study, and the rules used to decide which studies made it into this review and which did not.

2.2 Conceptual Framework

2.2.1 Concept of Attendance Management

Attendance management is how an institution records, stores, and uses information about whether a person was present for a scheduled activity. In a university, this usually means confirming a student attended a lecture, tutorial, or practical, and using that record later to decide if the student can sit for an exam. Akinduyite, Adetunmbi, Olabode and Ibidunmoye (2013) split attendance management into two groups. Conventional methods include the time sheet, the attendance register, and the time clock, none of which need any equipment beyond paper or a simple device. Automated methods include barcode systems, magnetic stripe cards, RFID, and biometric systems, each of which replaces manual recording with a machine reading some form of identification instead.

This split matters because it shows that a biometric attendance system is not something completely new, it is simply the next step after barcode and card-based systems that came before

it. What every version shares, manual or automated, is one basic requirement: the record has to actually show who was present, or the whole exercise is pointless. In Nigerian universities, this record is tied directly to whether a student can sit an exam, which is why attendance is taken seriously enough that institutions keep trying to find a better way to record it.

In the Nigerian university context, the stakes attached to attendance records go beyond administrative tidiness. Under the National Universities Commission minimum standards for most undergraduate programmes, a student who falls below 75 percent attendance in any course is barred from sitting that course's examination, regardless of how well they have done in tests or assignments during the semester. This means that every time a student is marked present when they were not, or marked absent when they were, the error can reach as far as that student's examination eligibility and, in the worst cases, their graduation timeline. A manual attendance system that cannot guarantee its own accuracy is, in this regulatory setting, not just inconvenient, it is a liability. This is the direct institutional context in which the studies reviewed in section 2.4 below were all built, even those that do not say so explicitly.

2.2.2 Concept of Biometrics

Biometrics means using a person's physical or behavioural traits to identify or confirm who they are. Physical traits include fingerprints, the face, the iris, and the shape of the hand. Behavioural traits include things like how someone types or walks. What makes a biometric trait different from a password or an ID card is that it is tied to the person's own body. It cannot be handed to someone else the way a card can, and it cannot be forgotten the way a password can, though this does not mean it is perfect, a point this chapter comes back to later.

Biometric systems usually work in one of two ways. Verification checks one fingerprint against one already-claimed identity, answering a narrow question: does this match the person it says it belongs to? Identification checks one fingerprint against an entire database to work out who it belongs to, with no claim made beforehand. Rahman, Rahman and Rahman (2018) point out that attendance systems usually use verification, since the student's identity, through a registration number or a login, is already known before the fingerprint is checked, which makes the process faster than checking against an entire database would be.

Several physical traits are reliable enough to serve as biometric identifiers, and the choice between them involves trade-offs in cost, accuracy, and practical fit. Fingerprints are the most widely used in attendance systems, partly because the hardware needed to read them is cheap and widely available, and partly because most people are already comfortable with fingerprint checks from their use in smartphones and similar devices. Iris recognition is arguably more accurate, since the pattern inside the eye is more complex than a fingerprint ridge arrangement and less prone to wear over time, but iris scanners cost significantly more than fingerprint sensors, which puts them out of reach for most Nigerian university budgets. Facial recognition requires a camera and consistent lighting, and as Mikailu, Faseke, Ishaya and Ahmed (2025) found, even a well-designed facial recognition system struggles when lecture halls have inconsistent or variable light levels. Voice recognition is not practical in a busy lecture theatre at all.

The choice of fingerprint recognition across most of the studies reviewed in this chapter therefore comes down to three practical facts: the sensors are cheap enough for a limited institutional budget, the matching process is fast enough not to hold up a class, and the technology is familiar enough that a student or lecturer who has never used a fingerprint scanner can usually figure it out within one or two attempts. These practical advantages do not make fingerprint recognition the best biometric modality in an absolute sense. They make it the most reasonable choice for the specific institutional setting most of the studies in this review are working in, and that is the sense in which this study uses it too.

2.2.3 Concept of Fingerprint Recognition

Fingerprint recognition uses the ridge pattern on a fingertip. These ridges form patterns generally grouped into loops, whorls, and arches, and within these patterns are smaller details called minutiae, points where a ridge ends or splits into two. Rahman, Rahman and Rahman (2018) describe two ways of matching fingerprints: pattern-based matching, which compares the overall loop, whorl, or arch shape, and minutiae-based matching, which compares the specific ridge endings and splits instead. Nearly every practical fingerprint attendance system, including the ones reviewed in this chapter, uses minutiae-based matching, since two fingers can share the same general shape while still having minutiae arranged in a way that is different enough to tell them apart.

A fingerprint system usually works in two steps. During enrollment, a sensor captures an image of the finger, and the system processes that image, often cleaning it up first, before pulling out the minutiae into a small digital file, which is what gets stored, not the raw image. During verification, this happens again on a new scan, and the new file is compared with the one already stored. If enough points line up closely enough, the system counts it as a match.

One aspect of fingerprint matching that is easy to miss is the threshold. Every matching algorithm requires a minimum similarity score before it counts two fingerprint samples as a match. Set that threshold high and the system becomes strict: it will rarely accept a wrong fingerprint, but it will also reject more real ones, because even two scans of the same finger taken seconds apart are not identical images. Set it low and the system becomes lenient: it accepts matches more easily, but the chance of accepting the wrong fingerprint goes up. For an attendance system in a university, where the main concern is preventing one person from signing in for another rather than protecting high-security access, a mid-range threshold that keeps the false rejection rate low enough to avoid disrupting a class is generally the better balance. The right threshold value will depend on the sensor model and the conditions in the room, which is one reason no single number applies across all the systems reviewed in this chapter.

2.2.4 Conceptual Framework Diagram

The diagram below brings these ideas together and shows, in simple terms, how this study moves from a problem to a proposed solution. Manual attendance recording creates the problems this study is concerned with, mainly impersonation, time loss, and poor record keeping. Fingerprint biometric technology is presented in the literature as a solution that works for these problems. This study takes that response and applies it to a proposed system design for Mountain Top University, which is expected, if it were built, to produce a more accurate and fraud-resistant attendance record than the manual process it would replace. The word *expected* is used deliberately here, since this study does not build or test the system; it only proposes a design based on what has worked in comparable settings.

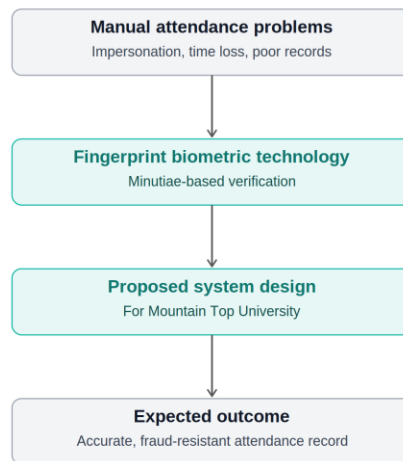


Figure 2.1: Conceptual framework showing the flow from problem to proposed system

2.3 Theoretical Framework

This study is built on minutiae-based fingerprint matching theory, the same idea explained in section 2.2.3, since it is the theory nearly every fingerprint attendance system in this chapter is actually built on. The theory rests on two ideas. First, the arrangement of minutiae on a finger is different enough between people that a match found between two samples almost certainly means they came from the same finger. Second, this arrangement stays stable enough over a person's life that a template captured once can still be matched years later.

How well a fingerprint system actually works is usually judged using three measures, all of which Akinduyite and colleagues (2013) apply in their own study. The false acceptance rate is the chance that someone who should be rejected gets accepted by mistake. The false rejection rate is the chance that someone who should be accepted gets rejected instead, often because of a poor scan. A third, less talked-about measure is the failure to enroll rate, which covers people who cannot be enrolled at all because their fingerprints are naturally faint or worn down, something Akinduyite and colleagues say affects around four percent of people, more often older individuals and manual workers whose fingertips have worn down over years of physical work.

Both ideas behind minutiae matching hold up well in normal conditions, which is why fingerprint recognition is still a common choice for identity checks generally. Neither idea is

perfect, though, and the failure-to-enroll rate above proves that. Minutiae can be temporarily hidden by cuts, moisture, or dirt, and can wear down for good after years of manual labour. A system designed with this theory in mind has to plan for this small margin of error rather than assume the technology is flawless, which is something this chapter comes back to in the critical analysis below.

2.3.1 Hardware and Sensor Technology in Fingerprint Attendance Systems

The theoretical ideas covered in sections 2.2 and 2.3 only become a working system when they are turned into physical hardware, and the choices made at that stage affect how the system performs in a real lecture hall. Hoo and Ibrahim (2019), in a twenty-five year survey of biometric attendance hardware published in the Journal of Sensors, found that three decisions matter most: the type of microcontroller used, the type of fingerprint sensor, and how attendance data is stored and shared once it has been captured.

Two sensor types appear across the literature. Optical sensors use a light source to photograph the ridge pattern on a finger. They are cheaper and straightforward to connect to a microcontroller, which is why they appear in most of the Nigerian university studies reviewed in section 2.4 below. The R305, R307, and AS608 modules used across those studies are all optical. Capacitive sensors use tiny electrical charges to build a fingerprint image instead, making them harder to trick with a photograph or a mould, but they cost more to buy. Hoo and Ibrahim (2019) note that optical sensors are more sensitive to a finger that is too dry or too wet, which connects directly to the failure-to-enroll problem raised in section 2.3.

Walia and Jain (2016) reviewed fingerprint attendance systems and grouped them by hardware approach into four types: standalone systems with local storage, GSM-enabled systems that send text alerts, IoT-connected systems that push data to a remote server, and more complex systems with full management dashboards. The pattern across the studies in section 2.4 below follows this trajectory. Earlier Nigerian systems, from 2013 to 2018, are mostly standalone. By 2021, IoT and cloud connectivity start appearing more regularly, and by 2024 the real-time cloud dashboard has become the default in newer studies. This matters for any implementation, because the choice between a standalone system and a cloud-connected one affects what staff

can do with attendance records, not just how they are captured, and that decision needs to be made before hardware is purchased rather than after.

On storage, Hoo and Ibrahim (2019) note that local memory cards work for small tests but create a practical problem in a real deployment: someone has to physically retrieve the card or connect a cable to export the data. Server-based storage removes that step entirely. This is one reason newer studies in section 2.4 lean toward cloud or server storage. The design proposed in Chapter Four takes the same position, treating the reporting side of the system as just as important as the fingerprint matching step.

The point where a system's false acceptance rate and its false rejection rate are equal is called the equal error rate, and it is sometimes used as a single summary measure of how accurate a fingerprint matching system is overall. A lower equal error rate means better overall performance across both types of error at once. In the studies reviewed in this chapter, the equal error rate is rarely reported directly. Akinduyite, Adetunmbi, Olabode and Ibidunmoye (2013) are an exception, giving both a zero percent false acceptance rate and a 2.56 percent false rejection rate under their test conditions, figures detailed enough that an equal error rate could be worked out, though they do not calculate it themselves. Most other studies report only one figure or neither, which makes cross-study comparison harder than it should be.

What this means in practice is that the threshold setting is not a technical detail to leave at a default value. It is a decision that directly affects whether the system is fair to students with worn or naturally faint fingerprints, whether it holds up a class when one person cannot be verified quickly, and whether it can be adjusted by the institution after deployment without needing the original developer to come back and change the code. Any future implementation at Mountain Top University would need to address this by setting the matching threshold deliberately for the specific sensor and room conditions being used, rather than leaving it at whatever default the sensor module ships with. That decision cannot be made at the design stage without knowing the exact hardware, but flagging it here means it cannot be overlooked at the implementation stage either.

One more practical detail from the theory is worth noting. The number of minutiae points that a system requires to match before it accepts two samples as the same finger varies by

implementation, and different systems set this differently depending on how strictly they want to guard against false acceptances. A lower required count means faster matching and fewer false rejections, but it raises the chance of accepting the wrong finger. A higher count does the opposite. In most of the Nigerian university studies reviewed in section 2.4 below, the minutiae count used is not reported at all, which makes it impossible to know whether differences in false rejection rates between studies reflect a genuine difference in sensor quality or simply a difference in how strict each implementation's matching parameters were set. This is another dimension of the standardisation gap noted in section 2.5 and another reason why comparing accuracy figures across studies requires caution.

A related hardware limitation, noted by Hoo and Ibrahim (2019) but absent from every implementation study reviewed in section 2.4, concerns what happens to stored fingerprint templates when the sensor itself is replaced or upgraded. Fingerprint templates are not raw images but processed numerical representations of ridge positions, and the processing algorithm used to generate those numbers is specific to the sensor module. When a sensor module is replaced with a different model, even a newer version of the same brand, the templates stored from the old sensor may not be compatible with the new one, because the two modules generate their numerical representations using slightly different methods. If that incompatibility exists, every enrolled user would need to re-enroll their fingerprint from scratch. In a university setting with hundreds or thousands of enrolled students, a forced re-enrollment is a significant operational cost, not just an inconvenience, and it is a cost that none of the systems reviewed in section 2.4 appear to have planned for. Any future implementation at Mountain Top University should confirm the template format used by the chosen sensor module before purchasing, and where possible choose a module with a standard template format that is compatible with alternative sensors, to reduce the risk of a forced re-enrollment when hardware eventually needs replacing.

Taken together, the three theoretical measures, false acceptance rate, false rejection rate, and failure to enroll rate, and the two practical tuning decisions, the similarity threshold and the minutiae count, describe a system that is not accurate or inaccurate in any fixed sense. It is a system whose performance is shaped by choices made at design and deployment stage. Any implementation that leaves the threshold and minutiae count at default values is effectively

making those decisions without acknowledging it. Noting them here, as decisions that need deliberate attention, is one way this review tries to be more practically useful to a future implementer than a literature review that stops at summarising accuracy figures and time comparisons.

2.4 Empirical Framework

This section reviews twenty-four studies that build on the concepts and theory explained above, grouped first around fingerprint systems built for Nigerian universities, then fingerprint systems built outside Nigeria, and finally systems that used a different technology altogether.

2.4.1 Fingerprint-Based Attendance Systems in Nigerian Universities

Akinduyite, Adetunmbi, Olabode and Ibidunmoye (2013) give the most detailed account, among the studies reviewed here, of a fingerprint attendance system built for a Nigerian university. Working at the Federal University of Technology, Akure, they built a system to deal with impersonation and what they call the ghost worker problem, and tested it on one hundred and seventeen real users, thirty staff and eighty-seven students, using a live scan. Their results are more specific than most in this area: a false acceptance rate of zero percent, a false rejection rate of 2.56 percent, an overall accuracy of 97.4 percent, and an average time of 4.29 seconds per person against 18.48 seconds for the manual method. Because this study gives its sample size, its error rates, and its method in enough detail to be checked, it works well as a benchmark for the less detailed studies below.

Elebede, Olusanya, Sodipo and Sam-Leeloo (2024) built a similar system about a decade later, presenting their work at an IEEE conference. Their study aimed directly at the unreliability of paper registers, and after testing it against the manual method it was meant to replace, they reported that the fingerprint system recorded attendance more consistently and left much less room for the kind of manipulation manual registers allow.

The IEEE NIGERCON conference setting is worth a brief note. Conference papers go through peer review, but review at conference level is generally less detailed than a journal review, and the work presented is often at an earlier stage of development than a full journal article would be. This does not make Elebede, Olusanya, Sodipo and Sam-Leeloo's (2024) findings unreliable. It

does mean they carry a little less evidential weight than the semester-long deployment study of Olasupo and colleagues (2022) or the large-sample controlled test of Akinduyite, Adetunmbi, Olabode and Ibidunmoye (2013), a distinction worth keeping in mind when reading across the studies in this section.

Olasupo and colleagues (2022) did something similar at Olabisi Onabanjo University, but instead of a short test, they used their system in real classes for a full semester. That difference matters, a system tested for a few days and one used continuously for months are answering two different questions. Lecturers who used the system all semester called it accurate, fast, and dependable, and students said it made it practically impossible for one person to sign in for another.

Shoewu, Makanjuola and Akinyemi (2018) worked on the same problem at Lagos State University's Epe campus, building a system called SAMSYS around a small microcontroller and a fingerprint reader. Their own account of the manual methods they had used for over fifteen years is worth repeating plainly, calling out names, passing round a sign-in sheet, and giving short quizzes after class, and all three had the same weakness: friends of absent students would write their names down or sit the quiz for them, something the authors say they noticed by comparing handwriting on submitted scripts. Once tested, their system cut the average time to mark attendance from 17.83 seconds under the manual method to 3.79 seconds with SAMSYS.

Obiora, Aladejare, Igbinosa, Fiemobebefa and Bamido (2025) built a similar system for the Department of Electrical and Electronics Engineering at Bells University of Technology. Their reason for building it was slightly different from the studies above, they wanted to cut down on proxy attendance and reduce the number of students who failed exams simply because their attendance was never properly recorded. Their system brought the time to mark attendance per student down from 22.25 seconds manually to 6.16 seconds with fingerprint verification.

Adejumobi, Adedoyin, Adenowo, Shoewu and Yussuff (2021) took a slightly different angle at Lagos State University, building a fingerprint system that automatically emails the compiled attendance record to relevant staff using standard email protocols. This extra step matters because it points to something the other studies above do not solve, even a perfectly accurate fingerprint system still needs someone to gather and share the resulting data unless that part is

automated too. This is worth keeping in mind for the design proposed later in this study, since stopping at fingerprint verification alone only solves half of the problem.

Onyishi and Igbinoba (2021), working at the University of Nigeria, Nsukka, built a similar fingerprint system but took an extra step before building the physical device, they first tested the full circuit design in Proteus, a simulation tool, to check that it worked correctly before constructing and testing the real thing. This is a detail worth noting for a design-only study such as this one, since it shows that a design can be checked for soundness even before a working prototype exists, which is close in spirit to what this study itself is trying to do.

Ana, Ekah and Oyo-Ita (2022) built an IoT-based fingerprint attendance system for Cross River University of Technology in Calabar using an ESP32 microcontroller. Unlike most of the systems reviewed above, which store attendance records on a local computer, this design uploads fingerprint data over Wi-Fi to a website in real time, so records are visible as students scan in without any manual file transfer needed. The ESP32 handles both the fingerprint module connection and the wireless link in one low-cost board, cutting out the separate GSM or Wi-Fi module that earlier Arduino-based designs required. CRUTECH is not an institution that appears often in this area of literature, so this study adds a direct reference point for a university not previously covered here.

Adedoyin, Shoewu, Yussuff, Adenowo, Shitta and Okedokun (2024) built a similar cloud-connected system at Lagos State University, replacing the standard Arduino with a NodeMCU board that has Wi-Fi built in and sending attendance records directly to a Firebase cloud database. A web dashboard built in React.js shows records updating in real time as students scan their fingers, which means a lecturer or administrator does not have to wait for the session to end or retrieve any files to see who attended. The system captured each fingerprint in roughly four seconds. This study comes from the same Lagos State University research group as Adejumobi, Adedoyin, Adenowo, Shoewu and Yussuff (2021) reviewed above, and the two papers together show that group moving from email-based reporting in 2021 to real-time cloud dashboards by 2024, which is a useful illustration of how fingerprint attendance systems continue to develop even within the same institution.

Adamu (2023) took the underlying idea further by combining fingerprint recognition with iris recognition in a single attendance system, reasoning that a system built around a single biometric feature has less room to fall back on if that one feature fails to read properly. Their system also automatically works out each student's attendance percentage against the same 75 percent rule mentioned in Chapter One, which ties this study directly to the same policy concern this project is built around.

Odulaja (2025), working at Tai Solarin University of Education, ran what is probably the most carefully structured comparison among the studies reviewed here. Attendance was taken manually for one semester and then with a fingerprint-based system, built around a small inference engine, for the next semester, with the same class of one hundred students across both. Manual manipulation of records dropped sharply in the second semester, correct identification reached 95 percent, and recorded attendance itself rose by 30 percent once the fingerprint system was in place, a rise the study puts down to students being less able to claim attendance without actually being present. Because this study compares the same group of people under both methods, one semester apart, rather than comparing two different groups, its result is one of the more convincing pieces of evidence in this chapter that a fingerprint system changes real behaviour, not just how attendance gets recorded on paper.

Oladiboye and colleagues (2026), working at Yaba College of Technology, built the most recent and most technically detailed system reviewed here, connecting a fingerprint sensor to a small microcontroller, a wireless link, and a cloud database, with a web page for staff to view records in real time. Tested with twenty students, the system kept a false rejection rate under 2 percent and a false acceptance rate under 0.5 percent, and completed a full attendance session in eight to ten minutes. What makes this study especially useful for this review is that the authors list their own study's limitations openly, a small number of students tested, sensitivity to moisture or cold on the finger, and a need for a stable network connection, rather than presenting the system as a finished, flawless product.

Odia, Oghogho and Ihedi-Okonkwo (2020), working at the University of Benin, built and tested a similar system, reporting that registering a new user took at most one minute and twenty seconds, while marking attendance for an already-enrolled person took under ten seconds. Their

system also produced a defaulter list automatically, flagging students who had fallen short of the attendance percentage needed for examination eligibility, without a staff member having to work this out by hand.

Adenekan, Isreal and Philip (2026) took the most recent and, in one respect, the most ambitious approach reviewed here, building an internet-connected fingerprint system at Moshood Abiola Polytechnic, Abeokuta, with the specific aim of studying how attendance recorded through the system related to students' academic performance over a semester. Where every other study in this section treats improved attendance recording as the end goal, this one treats it as a starting point for asking a further question, whether more accurate attendance data actually connects to how well students perform, a question none of the other studies reviewed in this chapter ask directly.

Looking at these fourteen studies together, spanning 2013 to 2026 and several different institutions, what stands out is how consistent the underlying problem is, impersonation, wasted class time, and unreliable manual records, even though the hardware and extra features differ from one study to the next. That this same pattern shows up across more than a decade and several institutions is itself a sign that the problem this study is addressing at Mountain Top University is not something made up for this project alone.

2.4.2 Fingerprint-Based Attendance Systems Outside Nigeria

Four studies here show that this same problem, and the same fingerprint-based answer to it, is not limited to Nigeria. Alhothaily, Alradaey, Oqbah and El-Kustaban (2015) built a fingerprint system at the University of Science and Technology in Sana'a, Yemen, driven by wasted lecture time and the difficulty of tracking whether students had met a minimum attendance rule, in their case 75 percent, close to the National Universities Commission rule discussed in Chapter One. Their system added something none of the Nigerian studies above had, a web and email interface that lets a student's guardian check attendance directly, stretching accountability beyond the school and into the home.

Asabere, Sekyere and Ofosu (2019) built a comparable fingerprint system at the University of Education, Winneba, in Ghana, using a low-cost microcontroller setup similar in spirit to several

of the Nigerian studies above, which suggests the affordability of this kind of system is not limited to any one country either. Chellaprabha, Dhilipkumar, Ajesh and Giridhar (2016) took a different route at SNS College of Engineering in Coimbatore, India, pairing fingerprint verification with a GSM module that sends a text message to a parent or guardian whenever a student is marked absent, an idea similar to Adejumobi and colleagues' automatic email reports, but aimed at parents rather than institutional staff.

The presence of studies from Ghana, Yemen, India, and Bangladesh in this review, each arriving at similar fingerprint attendance designs using similar low-cost hardware, is worth underlining. It suggests that the practical approach documented across the Nigerian studies in section 2.4.1 is not shaped purely by Nigerian institutional conditions. It reflects a broader pattern in university settings where budgets are limited, class sizes are large, and a manual attendance process is causing the same documented problems regardless of which country the university is in. That consistency across countries adds weight to the case for a fingerprint system at Mountain Top University, since the evidence base is not limited to one national context or one regulatory environment.

Rahman, Rahman and Rahman (2018) built a similar system at Jatiya Kabi Kazi Nazrul Islam University in Bangladesh, and their paper explains, in more technical detail than any other reviewed here, how minutiae-based matching is actually carried out, including the image clean-up steps done before minutiae are pulled out. What makes their study especially useful for this review, though, is that the authors admit its limits themselves, they say plainly that some users cannot be enrolled because of poor fingerprint quality, that fingerprints can change slightly with time, and that people may need to be re-enrolled later as a result. This is not a flaw pointed out by an outside reader, it is something the original authors say about their own work, which makes it a solid, trustworthy point to build on in the critical analysis below.

Hoo and Ibrahim (2019), from Universiti Sains Malaysia, did not build a new system but reviewed twenty-five years of biometric attendance hardware in education in a paper published in the *Journal of Sensors*. Their review is the most comprehensive hardware survey in this area and covers microcontrollers, sensor types, communication channels, and storage across dozens of published systems. On sensors, they confirmed that optical modules are the most common choice

in low-cost systems but are more sensitive to wet or dry fingers than capacitive alternatives. On storage, they found that server-based options have become preferred because staff can view records from any device without physically retrieving a memory card. Both findings tie directly to the sensor and storage discussion in section 2.3.1 above and help explain why the most recent Nigerian studies in section 2.4.1 look different from the earlier ones.

Patil, Khachane and Purohit (2016), at Vidyalankar Institute of Technology in Mumbai, India, designed a fingerprint attendance system that uses ZigBee wireless communication to link the classroom scanner to a central workstation. ZigBee is a short-range, low-power wireless protocol with a range of roughly thirty metres. It uses less electricity than Wi-Fi and costs less to run, but it cannot reach a remote server without an extra relay device. Their system used an ARM7 processor, stored up to one hundred fingerprint IDs in flash memory, and transferred data reliably within its range during testing. The study is notable because it addresses something most of the other papers in this review do not: where to physically put the scanner. Most systems assume it sits next to a computer, which is manageable in a small room but difficult in a large lecture hall. ZigBee is one answer to that placement problem, though the thirty-metre limit means a larger campus would need multiple units or a different wireless approach.

2.4.3 Alternative Automated Attendance Approaches

Not everyone working on this problem has chosen fingerprint recognition. Olanipekun and Boyinbode (2015) built an RFID system instead, giving each student a tag that a reader picks up automatically. Their study matters less for its technical results and more for how clearly it lists the specific problems with manual attendance, records that are hard to keep intact over a semester, time lost to the physical act of signing in, gaps in the paperwork, and students signing in for absent classmates, all points echoed by several of the fingerprint studies above. Their own solution, though, brings its own weakness, a tag, unlike a fingerprint, can be handed to someone else.

Mikailu, Faseke, Ishaya and Ahmed (2025) went a different way again, building a facial recognition system for Nigerian universities. Their study adds something the older studies leave out, they connect proxy attendance and extra work for staff directly to the National Universities Commission's minimum attendance rule for exam eligibility, spelling out a link the fingerprint

studies above mostly leave unspoken. Their system does not use fingerprints at all, though, which limits how directly it can be compared to the fingerprint studies in sections 2.4.1 and 2.4.2, even though it is solving the same underlying problem.

Badmus, Odekunle and Oyewobi (2021), working across the University of Ibadan and Federal University Oye-Ekiti, combined fingerprint recognition with RFID in a single attendance check. Their reasoning was that a system built around one technology alone has no fallback if that technology fails for a particular student. In their design a student presents both a fingerprint and an RFID tag, and the system marks attendance only when both inputs match the same registered person. This answers the fallback question differently from Adamu (2023) reviewed in section 2.4.1, whose dual-biometric approach pairs fingerprint recognition with iris recognition instead. The fingerprint-plus-RFID option here is cheaper to deploy, but it reintroduces the card-loss problem that biometric-only systems were meant to remove. Neither study resolves that tension cleanly, and both say so plainly, which makes them more honest about their limits than most of the literature in section 2.4.1.

What Badmus, Odekunle and Oyewobi (2021) add that most other studies do not is a direct answer to what happens when a student forgets their RFID card on a given day. Their system requires both inputs to record attendance, so a student without their card simply cannot be marked present that day, regardless of whether their fingerprint is in the database. This sounds like a design limitation, and it is, but it is also an honest acknowledgment that every system has a fallback failure point. Most of the purely fingerprint-based studies reviewed in section 2.4.1 do not address their own equivalent failure point, which is the student whose fingerprint cannot be read. Badmus, Odekunle and Oyewobi (2021) are unusual in this literature for making the failure condition explicit at the design stage rather than discovering it during testing.

2.4.4 Summary of Reviewed Studies

Three things stand out when reading across the twenty-four studies reviewed in this chapter. First, the core hardware, a fingerprint sensor connected to a microcontroller, has stayed the same while the surrounding architecture has moved from local-only to cloud-connected. Second, every study that compares fingerprint and manual methods reports a faster result with fingerprinting, usually by a factor of three to five. Third, accuracy figures cluster between 94 and 97 percent

across the studies that report them, suggesting a ceiling at this hardware level rather than a trend toward improvement.

The table below summarises the key details of the Nigerian fingerprint studies reviewed in section 2.4.1, showing the institution, year, technology, and the standout feature or finding for each. Studies reviewed in sections 2.4.2 and 2.4.3 are not included since they serve a comparative rather than a primary role in this review.

Author(s) and Year	Institution	Key Feature or Finding
Akinduyite, Adetunmbi, Olabode and Ibidunmoye (2013)	FUTA, Akure	97.4% accuracy; 0% FAR; 4.29s vs 18.48s; 117 users tested
Shoewu, Makanjuola and Akinyemi (2018)	Lagos State University	SAMSYS; 3.79s vs 17.83s; microcontroller with Bluetooth
Odia, Oghogho and Ihedi-Okonkwo (2020)	University of Benin	Auto-generates defaulter list; under 10s per scan
Adejumobi, Adedoyin, Adenowo, Shoewu and Yussuff (2021)	Lagos State University	First in review to automate email reporting to staff
Onyishi and Igbinoba (2021)	University of Nigeria, Nsukka	Circuit pre-tested in Proteus simulation before build
Olasupo and colleagues (2022)	Olabisi Onabanjo University	Full-semester real deployment; lecturers and students both reported positive outcomes
Ana, Ekah and Oyo-Ita (2022)	CRUTECH, Calabar	ESP32 with real-time Wi-Fi upload; underrepresented institution in this literature
Adamu (2023)	Not specified	Dual biometric — fingerprint plus iris; auto-calculates 75% eligibility
Adedoyin, Shoewu, Yussuff, Adenowo, Shitta and Okedokun (2024)	Lagos State University	NodeMCU with Firebase cloud; React.js real-time dashboard
Elebede, Olusanya, Sodipo and Sam-Leeloo (2024)	IEEE NIGERCON	Conference paper; consistent recording; less manipulation than paper
Obiora, Aladejare, Igbinosa,	Bells University of	6.16s vs 22.25s; reduces proxy and unrecorded

Fiemobebefa and Bamido (2025)	Technology	absence
Odulaja (2025)	Tai Solarin University of Education	Two-semester same-group comparison; 30% attendance rise; 95% accuracy
Oladiboye and colleagues (2026)	Yaba College of Technology	Most recent; FRR under 2%; FAR under 0.5%; openly lists limitations
Adenekan, Isreal and Philip (2026)	Moshood Abiola Polytechnic	Only study linking attendance data to academic performance

Table 2.1: Summary of Nigerian fingerprint attendance studies reviewed in section 2.4.1

2.4.5 Implementation Challenges in the Nigerian University Context

Several of the studies reviewed in sections 2.4.1 to 2.4.3 mention practical difficulties that come up when a fingerprint attendance system moves from a controlled test into a real institution. These are worth pulling together in one place, because a design that does not account for them risks being technically sound but difficult to use once it is actually deployed in a lecture hall.

The most commonly mentioned difficulty is power supply. Nigeria's electricity grid is widely documented as unreliable, with outages affecting institutions across the country at irregular intervals. Any electronic system deployed in a Nigerian lecture hall has to account for this, and a fingerprint attendance system is no exception. A system that stops working every time the power goes out does not solve the attendance problem, it adds a different one. Any future implementation at Mountain Top University would need to plan for this from the start, whether by including a local storage option so that records captured before an outage are not lost, by using a battery backup for the scanner, or both.

A second difficulty is network reliability. Cloud-connected systems like those built by Ana, Ekah and Oyo-Ita (2022), Adedoyin, Shoewu, Yussuff, Adenowo, Shitta and Okedokun (2024), and Oladiboye and colleagues (2026) depend on a working internet connection to push records to a server in real time. Where that connection is slow or drops out, real-time reporting either fails or introduces delays that make the live dashboard less useful than a simple local record would have

been. None of these studies say what happened to their system during a network outage, which is a gap worth noting for any institution considering a cloud-connected implementation.

A third difficulty is environmental. Optical fingerprint sensors are sensitive to humidity, dust, and repeated use. In a crowded lecture hall in a warm climate, a sensor can collect moisture and oils from many fingers within a single session, which degrades the image it captures and raises the false rejection rate over the course of the class. Hoo and Ibrahim (2019) flag this as a consistent hardware limitation across the biometric attendance literature. Regular cleaning of the sensor surface is a maintenance step none of the reviewed studies mention explicitly, but it is likely necessary in regular deployment. A university implementing a fingerprint attendance system needs to factor sensor maintenance into its planning from the start, not treat it as something to deal with when problems arise.

2.5 Critical Analysis of the Reviewed Literature

Looking at these twenty-four studies side by side, rather than one at a time, brings out some weak points that are easy to miss when each paper stands alone. The first is about scale and how long each system was actually tested. Akinduyite and colleagues (2013) tested on 117 users, one of the bigger samples among the studies reviewed here, while Oladiboye and colleagues (2026) tested on just twenty, a small enough number that the authors themselves say a larger trial is needed before the results can be fully trusted. Olasupo and colleagues (2022) and Odulaja (2025) stand out as the only two who describe a full semester of ordinary use rather than a short test set up by the researchers themselves, with Odulaja's design being especially convincing since it compares the same one hundred students under both the manual and the fingerprint method rather than comparing two different groups. Even so, a system that works well for a few hundred checks over a few days, or even one semester at one institution, has not necessarily proven it will work the same way across a whole university, with several colleges and a much wider range of lecture halls, network conditions, and office staff, over several years.

A second weak point is about where this literature actually comes from. While putting this review together, several papers found through search engines turned out, once read closely, to be the same underlying study posted under different titles on different repository sites, sharing the same abstract and the same numbers despite looking like separate publications. This is not a

criticism of any one author, since it seems to come from how repository sites index and mirror content rather than anything researchers did on purpose, but it is still a real problem when working with this kind of literature, a figure that looks confirmed by several sources may actually trace back to just one. Every source used in this chapter was checked individually against its real publishing venue and, where one existed, its DOI, before being included here, a process explained further in section 2.8.

A third weak point, and maybe the most important one for a study like this that only produces a design rather than a working system, is that almost none of this literature says what happens when fingerprint matching fails. Two of the studies address this directly. Rahman, Rahman and Rahman (2018) admit that some users cannot be enrolled at all and that fingerprints can change over time, and Oladiboye and colleagues (2026) separately note that moisture and cold can affect a scan, matching the failure-to-enroll idea covered in section 2.3. That two studies published years apart, in different countries, arrive at the same honest admission on their own suggests this is a real, recurring limitation rather than a one-off problem. None of the remaining Nigerian studies in section 2.4.1 say what their systems would do for a student or staff member who cannot be reliably enrolled, whether because of a manual job, an injury, or just a naturally faint print, a gap this study treats as something the design in Chapter Four needs to plan for, not a small detail to skip. Adamu's (2023) choice to combine fingerprint and iris recognition in one system points to one possible answer, giving a second biometric feature to fall back on, though even that study does not say what happens if a person struggles with both.

A fourth point is about how these studies explain why the problem matters in the first place. Only Mikailu and colleagues (2025), and to some extent Alhothaily and colleagues (2015) and Adamu (2023), directly tie attendance recording to a specific policy that decides exam eligibility. The rest, including most of the Nigerian fingerprint studies in section 2.4.1, mostly talk about impersonation and extra work for staff without saying plainly why these issues matter so much to the school, leaving the link to exam eligibility to be assumed rather than stated. This study states that link plainly in Chapter One and treats it as central to why the proposed system matters.

Lastly, the extra features that set these studies apart from one another, automatic email reports in Adejumobi and colleagues (2021), guardian web access in Alhothaily and colleagues (2015), text

message alerts to parents in Chellaprabha and colleagues (2016), simulation-based testing in Onyishi and Igbinoba (2021), and cloud-based real-time reporting in Oladiboye and colleagues (2026), are each presented as a standalone addition in their own paper rather than compared against each other directly. No single study reviewed here brings together automatic reporting, notification to guardians or staff, and a clear plan for enrollment failure in one design, and that gap is part of what this study responds to.

A fifth point, and one that is almost entirely absent from this literature, is cost. Hoo and Ibrahim (2019) note that a basic Arduino-based fingerprint system can be assembled for well under a hundred US dollars in parts, which probably explains why this type of setup dominates the Nigerian studies reviewed here. But none of those studies give a breakdown of what their system actually cost to build, which makes it impossible to compare them on value or to estimate what a version for Mountain Top University would need in terms of budget. Walia and Jain (2016) noted the same gap in an earlier review, and little has changed. A literature that reports accuracy and speed but not cost gives only part of the picture that a real institution would need before committing to build anything. Any implementation working from the design proposed in Chapter Four would be well advised to stay within the hardware options that appear most often in the low-cost Nigerian studies above rather than defaulting to more expensive alternatives that have no track record in this context.

A sixth point is hardware standardisation. The studies reviewed here use several different fingerprint modules, the R305, R307, and AS608, and several different microcontrollers, Arduino Uno, Arduino Mega, NodeMCU, and ESP32, without explaining why one was chosen over another beyond general references to cost and availability. Hoo and Ibrahim (2019) flag this as a gap across the biometric attendance literature generally: a system using one sensor may report a different false rejection rate from one using a different sensor, not because the matching logic differs but because the sensors capture fingerprint images at different resolutions. None of the papers reviewed here account for that difference when reporting their accuracy figures. A future implementation at Mountain Top University would need to test its specific hardware in the institution's own lecture halls rather than assuming the numbers from these studies would transfer exactly.

A seventh point, and one that none of the twenty-four studies in this review addresses at all, is student data privacy. A fingerprint template is biometric data. Unlike a password, it cannot be changed if it is compromised. Once a university stores a student's fingerprint template, that data is associated with that student for as long as it exists, and a breach of the system would expose something the student can never alter. Nigeria's Data Protection Act of 2023 places obligations on organisations that collect and store personal data, including biometric data, around consent, storage limits, and secure disposal. None of the Nigerian studies reviewed here mention this regulatory context. Most were written before the 2023 Act came into force, but the earlier Nigeria Data Protection Regulation of 2019 carried similar principles. An institution implementing any of these systems today, including one based on the design proposed in this study, would need to address data privacy requirements explicitly, not as an afterthought after the system is built. This includes informing students of what data is being collected and why, setting a retention policy that removes templates when a student is no longer enrolled, and ensuring the database is secured against unauthorised access. The absence of this discussion from the existing literature is a gap this review flags rather than resolves, since the legal requirements are outside the scope of this study's design work. But flagging it here means it cannot be overlooked at the implementation stage.

2.6 Research Gap

Even with all this work, no published study proposes a fingerprint biometric attendance system specifically for Mountain Top University, at the design stage or beyond it. This is not a strange gap, since Mountain Top University is a fairly young institution and has not come up as a case study in this literature the way Olabisi Onabanjo University, Bells University of Technology, or Lagos State University have. More importantly, every fingerprint study in this chapter, Nigerian or otherwise, is a build-it-and-test-it study, done by a researcher who had direct access to the institution and built a working system to try out, even the most recent one reviewed here, Oladiboye and colleagues (2026). None of them approach the problem the way this study does, working entirely from existing literature to produce a design for an institution the researcher did not have direct access to, and saying so plainly rather than working around it. A second, narrower gap is about enrollment failure, as noted in section 2.5, only two of the twenty-four studies reviewed here talk about what happens when a fingerprint cannot be captured properly,

and none of them suggest a specific backup plan for this within a Nigerian university. This study fills both gaps, a design proposed for an institution not yet covered in this area of research, shaped by a review that treats enrollment failure as something to plan for rather than something to ignore.

A third gap concerns how fingerprint attendance records connect to the wider institutional systems that actually use those records. None of the twenty-four studies reviewed here describe a fingerprint attendance system that feeds directly into a university's existing student information system or examination eligibility database. Every system reviewed produces its own attendance records, but the step of transferring those records into the system that determines whether a student can sit an exam is left unaddressed. Adejumobi, Adedoyin, Adenowo, Shoewu and Yussuff (2021) come closest by automating email reports to staff, and Adamu (2023) calculates each student's attendance percentage automatically, but neither connects to an existing institutional database. For Mountain Top University, where attendance records have to feed into examination eligibility decisions governed by the NUC's 75 percent rule, that integration gap is not a minor technical detail. Any future implementation at Mountain Top University would need to treat this integration as a core requirement from the start rather than something to add on later, since the attendance record is only as useful as the system that decides what to do with it.

A fourth gap is what might be called the post-implementation gap. Every study reviewed here reports on the initial deployment or testing phase of its system, but none follows up on whether the system was still in use six months or a year later, whether it was maintained, updated, or replaced, or whether the behaviour changes it produced in students and staff lasted beyond the study period. Odulaja (2025) comes closest to addressing longevity by comparing the same group of students across two full semesters, but even that study ends at the semester boundary. A literature that only reports initial results cannot tell a future implementer whether the system they are about to build will still be working and still be used in three years' time, and that is a practical question any institution needs answered before committing to an implementation.

2.7 Significance of the Literature Review to This Study

This review does more than show that a gap exists. Each of its main parts answers one of the research questions from Chapter One directly. The conceptual and theoretical sections answer

the first question, explaining what fingerprint recognition is and how well it works, which supports the design decisions made in Chapter Four. The review of related studies in section 2.4 answers the second question, showing exactly how fingerprint biometric attendance systems have been built and used in Nigerian and international university settings that are similar to this one. The critical analysis in section 2.5 feeds directly into the third question, since the weak points found there, especially around automatic reporting, enrollment failure, and the link to exam eligibility, are treated as things the design has to account for, not side notes, once the proposed system's structure is set out later in this study. In this sense, the literature reviewed here is not just background reading kept apart from the rest of the project, it is the evidence the proposed design actually stands on.

There is also a methodological significance to this review that is worth stating plainly. Checking twenty-four sources individually against their actual publishing venues, rather than relying on secondary listings or repository copies, took considerably more time than a standard search-and-cite approach would have. But it revealed at least three sources that appeared to be separate publications but turned out to be mirrored or near-identical copies of the same underlying work posted on different repository sites, and it excluded one further paper whose reference list contained citations entirely unrelated to its claimed subject, a sign of a low-quality or fabricated publication. For any future researcher working in this same area, this review's source-checking approach offers a documented example of why individual verification matters in a field where preprint and repository mirroring is common. It also justifies the scope section below, since the inclusion rules applied here are more specific than a general keyword search would have been.

The significance of explicitly identifying and excluding duplicates goes beyond keeping this review honest. It affects the evidential weight of the conclusions the review can support. A literature review that appears to draw on twenty-four independent sources but where three or four of those are actually the same study in different wrappers is a review that overstates how settled the evidence is. The critical analysis in section 2.5, for instance, identifies hardware inconsistency as a gap across the reviewed studies. That observation would carry less force if some of the studies showing inconsistent hardware choices turned out to be the same study counted twice. By verifying each source individually before including it, this review can claim

more confidently that the patterns it identifies, and the gaps it flags, are genuine features of the literature as a body of independent work, not artefacts of the search process.

2.8 Scope of the Review

This review only included studies that met a few specific conditions. Each one had to focus on fingerprint recognition, or a clearly comparable automated method, applied to attendance in an educational setting, and had to be traceable to a real, checkable place, a journal, conference, or accredited repository, with a working publication record or DOI. Nigerian university studies were given priority, given the shared National Universities Commission context explained in Chapter One, though four international fingerprint studies and two studies using other technology were included on purpose, since they add a comparison this review would otherwise be missing, not because they matched the Nigerian setting exactly.

A few types of material were left out on purpose. Sources that could not be checked as real, separate publications, including the duplicate and mirrored postings found while preparing this review, were dropped once spotted, even where they looked relevant at first glance. Company blogs, adverts, and other non-academic material were left out completely. Studies applying fingerprint or other biometric technology to things unrelated to attendance, such as banking or border checks, were also left out unless a specific technical point from them was already covered more directly by the attendance-focused literature used here. One further source, a survey of staff opinions on different biometric technologies at a private company, was found during this review but left out of the final list since it was not about a university setting and did not test a working system the way the rest of this chapter's sources do. Finally, sources whose authors could not be confirmed on the page where they were found were left out rather than cited on a guess, in keeping with the same checking standard used throughout this study.

One further decision worth recording here is the treatment of review articles and hardware surveys. Walia and Jain (2016) and Hoo and Ibrahim (2019) are not primary implementation studies. They do not build or test a fingerprint system. Both were included in this review because they each contribute something the implementation studies do not provide on their own. Walia and Jain (2016) offer a classification of fingerprint attendance systems by hardware type that helps make sense of the range of designs in section 2.4.1. Hoo and Ibrahim (2019) provide the

most detailed account in this literature of why particular hardware choices are made and what their practical consequences are, connecting sensor type to reliability in a way no implementation paper in this review does. Including two such sources alongside twenty-two implementation or near-implementation studies keeps the balance of this review firmly on the side of empirical work while still drawing on the broader perspective that survey papers provide.

2.9 Summary

This chapter has covered the ideas behind attendance management, biometrics, and fingerprint recognition, set out the minutiae-based matching theory this study relies on, examined the hardware and sensor choices that shape how fingerprint systems work in practice, and looked closely at twenty-four studies on fingerprint and other automated attendance systems, fifteen built for Nigerian universities, six built internationally, and three using a different technology or approach. The critical analysis found six weak points across this literature: small testing scale, duplicated sources, the failure-to-enroll gap, the missing link to exam eligibility policy, the absence of cost data, and inconsistent hardware choices across studies. All six are treated as things to design around rather than reasons to doubt the case for a fingerprint biometric attendance system at Mountain Top University. No existing study covers Mountain Top University specifically, and this study is meant to fill that gap through the design work carried out in the chapters that follow.

What the chapter as a whole establishes is not just that a fingerprint biometric attendance system is a reasonable choice for Mountain Top University. It establishes the specific shape of the problem the system needs to solve, the weak points in the existing literature that its design should address, and the hardware and connectivity decisions that will affect whether it works in practice. Each of those findings feeds directly into Chapter Three's methodology and Chapter Four's design, making this review something the rest of the study is built on rather than background reading that could have been left out.

CHAPTER THREE

RESEARCH METHODOLOGY

3.1 Introduction

This chapter explains how this study was carried out. It states the research design used, where the data for this study came from, how the current manual attendance process was studied through literature rather than direct observation, what the proposed system needs to do, and which system design method was used to plan it. Every choice explained in this chapter follows from one fact stated plainly in Chapter One: this study did not have direct access to Mountain Top University, so it relies on secondary data rather than data collected first-hand.

3.2 Research Design

This study uses a design and development research approach. This type of research does not test a hypothesis using collected data, and it does not survey a group of people to describe their views. Instead, it sets out to produce a design, in this case the design of a fingerprint biometric attendance system, based on what has already been tested and reported elsewhere. The output of this kind of study is not a set of statistics but a system design, supported by diagrams, that could guide a future implementation.

This approach fits the present study for a simple reason: there is no data being statistically tested, since no primary data was collected from Mountain Top University. What this study relies on instead is secondary data, meaning information already published by other researchers, which is explained further in the next section. A design and development approach is built for exactly this situation, where the researcher's job is to produce a sound design from existing evidence rather than to gather new evidence directly.

3.3 Sources of Data

All data used in this study is secondary. Two kinds of secondary sources were used. The first kind is the empirical literature reviewed in Chapter Two, twenty-four studies on fingerprint and other automated attendance systems, checked one by one to confirm they are real and distinct before being used. These studies were the basis for understanding both the weaknesses of manual

attendance recording and the way other researchers have addressed those weaknesses with fingerprint technology.

The second kind is general reference material on systems analysis and design, used to choose and explain the design method applied in section 3.7 below and carried into Chapter Four. No data was collected directly from staff or students at Mountain Top University, and no questionnaire, interview, or observation was carried out at the institution, for the reason already stated in Chapter One, direct access was not available at the time of this study.

Checking each source individually meant confirming that the journal or conference named in the citation actually existed, that the paper appeared in that venue, and where a digital object identifier was available, that it resolved to the correct record. Several sources found through initial keyword searches were excluded after this check. Some turned out to be the same underlying paper posted under slightly different titles on different repository sites, sharing the same abstract and the same results despite appearing to be separate studies. One was excluded because its reference list contained citations with no traceable connection to its claimed subject. The twenty-four studies that made it into Chapter Two are the ones that passed this individual check, not simply those that turned up in a search.

3.4 Analysis of the Existing System

Since this study could not observe Mountain Top University's own attendance process directly, the existing system described here is based on the manual attendance process documented across the literature reviewed in Chapter Two, which is common to Nigerian universities generally. In this process, a lecturer or a class representative records attendance in one of a few ways, calling out each student's name and marking them present or absent, or passing round a sheet of paper for each student to sign. Shoewu, Makanjuola and Akinyemi (2018) describe a third method some lecturers use, a short quiz given at the end of class, with the student's answer script standing in as proof of attendance.

Whichever method is used, the record produced is written by hand and stored on paper, usually kept by the lecturer or submitted to a departmental office at the end of the semester. At that point, someone has to go through every sheet or register and work out each student's attendance

percentage by hand, a process that has to happen before a final decision can be made on whether each student is allowed to sit for the semester's examinations.

The diagram below sets out this process as a simple flow, from the start of a class to the point where an examination eligibility decision is made. Laying it out this way makes it easier to see exactly where the weaknesses discussed in the next section come from, since each step shown here is a point where a record can be falsified, lost, or delayed.

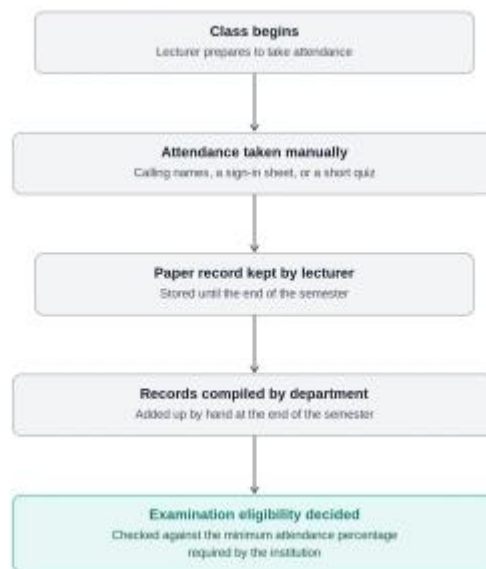


Figure 3.1: The manual attendance process this study responds to

3.5 Weaknesses of the Existing System

Chapters One and Two already set out the specific weaknesses of this manual process in detail, so they are only summarised briefly here rather than repeated in full. The most serious weakness is that a student can sign in, answer a quiz, or be called present by a classmate on their behalf, without the lecturer ever noticing. Beyond this, the process wastes class time, produces records that are often incomplete or hard to read, and places a heavy manual workload on whoever has to add up attendance figures at the end of the semester. These weaknesses are the direct reason a fingerprint biometric attendance system is being proposed in this study.

3.6 Requirements of the Proposed System

3.6.1 Functional Requirements

Functional requirements describe what the proposed system must be able to do. Based on the weaknesses identified above and the features found across the literature reviewed in Chapter Two, the proposed system should be able to:

- i. enroll a student or staff member by capturing their fingerprint along with their basic identifying details;
- ii. verify a person's fingerprint against their stored record before marking them present for a class;
- iii. record the date, time, and course for each attendance entry;
- iv. generate an attendance report for a student, a course, or a department on request;
- v. flag a case where a fingerprint cannot be matched, so a staff member can look into it, rather than silently marking the person absent;
- vi. allow an administrator to register new courses and assign lecturers to them.

The fifth requirement follows directly from a gap identified in Chapter Two: most of the fingerprint attendance studies reviewed did not explain what their system does when a fingerprint cannot be read properly. This study treats that as something the design has to plan for, not something to leave out.

3.6.2 Non-Functional Requirements

Non-functional requirements describe how well the system should perform its functions, rather than what those functions are. The proposed system should be:

- i. accurate, so that a genuine attendance record is rarely rejected and a false one is rarely accepted;
- ii. fast enough that verifying one person's fingerprint does not take noticeably longer than the few seconds reported in similar studies reviewed in Chapter Two;

- iii. easy enough to use that a lecturer or student needs little to no training before using it;
- iv. secure, so that stored fingerprint data and attendance records cannot be viewed or changed by someone without permission to do so.

3.7 System Design Methodology

This study adopts Object-Oriented Analysis and Design, commonly shortened to OOAD, as the method used to design the proposed system in Chapter Four. Booch (2007) describes this approach as one that models a system as a set of objects, each with its own data and behaviour, rather than as a single block of steps to be carried out in order. A student, a fingerprint record, and an attendance entry can each be treated as an object in this sense, with clear relationships between them.

Two reasons support this choice. First, OOAD is built around diagrams, particularly use case diagrams and entity relationship diagrams, which are exactly the tools this study uses in Chapter Four to describe how the proposed system would work and how its data would be organised. Second, because this study is design-only and does not involve writing or testing code, a method centred on clear diagrams and defined relationships between parts of the system is more useful here than a method centred on the order in which code gets written.

Using OOAD, the design process in Chapter Four follows two broad steps. The first step, analysis, identifies the objects the system needs, such as the student, the staff member, the course, and the attendance record, and works out how they relate to one another. The second step, design, turns this understanding into the actual diagrams presented in Chapter Four, showing how a user interacts with the system and how its data is structured.

Other approaches were considered before settling on OOAD. The Waterfall model, one of the earliest and most widely taught approaches to software development, treats a project as a sequence of fixed phases, requirements, design, implementation, testing, and maintenance, with each phase completed before the next begins. This works well for a project that will actually be built and tested, but it assumes an implementation phase that this study does not carry out, which makes it a poor match for a design-only project such as this one. Agile approaches, which break a project into short, repeated cycles of building and testing with continuous feedback, run into

the same problem, they assume working software is being built and revised along the way, something this study does not do either.

OOAD does not carry this assumption. It is concerned with understanding a system's structure, its objects, their data, and their relationships, rather than with the sequence of building and testing that structure. This makes it usable at the design stage alone, without needing to reach the implementation or testing phases that Waterfall and Agile are both built around. This is the main reason OOAD was chosen over these alternatives for this study, not because it is a better methodology in general, but because it fits a project that stops at the design of a system rather than carrying it through to a working product.

One further point connects the methodology directly to the diagrams in Chapter Four. The use case diagram and entity relationship diagram presented there are both drawn from the Unified Modeling Language, commonly known as UML, which is the standard visual language used within the object-oriented tradition to describe software systems. Booch, whose work on OOAD this study cites above, was also one of the three original authors of UML, so the connection between the methodology choice and the diagramming tools used in Chapter Four is not coincidental. Choosing OOAD as the methodology means UML diagrams are the natural output, and that is exactly what Chapter Four produces.

3.8 Tools and Development Environment

Although this study does not implement the proposed system, it is useful to indicate the kind of tools a future implementation could reasonably use, based on what has worked in the literature reviewed in Chapter Two. For the fingerprint hardware itself, an optical fingerprint sensor such as the R305 or AS608 module would be suitable, the same general type of sensor used in several of the studies reviewed, including Rahman, Rahman and Rahman (2018) and Shoewu, Makanjuola and Akinyemi (2018).

For the underlying programming language and database, a language such as C# or Python, paired with a relational database such as MySQL or Microsoft SQL Server, would be consistent with the approach taken by Akinduyite and colleagues (2013) and Rahman and colleagues (2018), both of which used a similar combination successfully. For a more modern deployment, Odia

and colleagues (2020) and Adenekan and colleagues (2026) show that a cloud-based database, accessed over the internet rather than stored only on a single local machine, is also a workable option, and would suit Mountain Top University if the institution wanted attendance data accessible from more than one office at a time.

None of these tool choices are fixed requirements of the design presented in Chapter Four, they are reasonable options consistent with what has already been shown to work elsewhere in the literature reviewed. A future implementation would still need to weigh cost, available expertise, and existing infrastructure before settling on one combination over another.

3.9 Summary

This chapter has explained that this study follows a design and development research approach, relying entirely on secondary data since direct access to Mountain Top University was not available. It has described the manual attendance process this study responds to, based on the literature reviewed in Chapter Two, summarised the weaknesses of that process, and set out the functional and non-functional requirements the proposed system needs to meet. It has also explained why Object-Oriented Analysis and Design was chosen over alternatives such as Waterfall and Agile, and outlined the kind of tools a future implementation could reasonably draw on. Together, these choices form the basis for the design work presented in the next chapter.

CHAPTER FOUR

DATA ANALYSIS AND PRESENTATION

4.1 Introduction

In keeping with the department's standard project format, this chapter serves as the system design equivalent of a data analysis chapter. It presents the proposed system architecture, diagrams, and database design in place of analyzed primary data, since this study is design-only and relies entirely on secondary sources.

This chapter presents the design of the proposed fingerprint biometric attendance system for Mountain Top University. It builds directly on the requirements set out in Chapter Three, using Object-Oriented Analysis and Design as explained there. The chapter covers the overall system architecture, the use case diagram showing who can do what within the system, the entity relationship diagram showing how the system's data is organised, a simple database design based on that diagram, and a description of how the system would work step by step, from enrollment through to a completed attendance record. As stated throughout this study, no part of this chapter has been built or tested. It is a design only, meant to guide a future implementation.

This chapter answers the third research question set out in Chapter One, what system architecture and design would be appropriate for a fingerprint biometric attendance system proposed for Mountain Top University. The first two research questions were answered in Chapter Two, through the literature review and the studies examined there. Everything presented from this point forward in the chapter is this study's answer to that third question.

4.2 System Architecture

The proposed system is made up of four main parts that work together. The enrollment part is where a student or staff member's fingerprint is captured, along with their basic details, and turned into a stored record. The verification part is what runs each time someone places their finger on a reader to be marked present, comparing the new scan against the record stored during enrollment. The database is where every user's details, fingerprint record, course information,

and attendance history are kept. The reporting part is what pulls information out of the database and presents it as a usable attendance report for a student, a course, or a department.

These four parts match the functional requirements already set out in Chapter Three. Enrollment and verification cover the first two requirements, the database supports the third requirement, recording the date, time, and course for each entry, and the reporting part covers the fourth requirement. The fifth requirement, flagging a case where a fingerprint cannot be matched, sits inside the verification part rather than as a separate module, since it is something that part of the system needs to check for every time it runs, not a separate process on its own. The sixth requirement, letting an administrator register courses and assign lecturers, is handled through the database as well, since it is really just another kind of record being added or changed.

4.3 Use Case Diagram

The diagram below shows the three types of people, or actors, who would use the proposed system, and what each one is able to do. A student verifies their own fingerprint to be marked present for a class. A lecturer can review a case where a fingerprint could not be matched, and can generate an attendance report for their own course. An administrator handles the setup side of the system, enrolling a new fingerprint record for a student or staff member, registering a new course, and assigning a lecturer to a course.

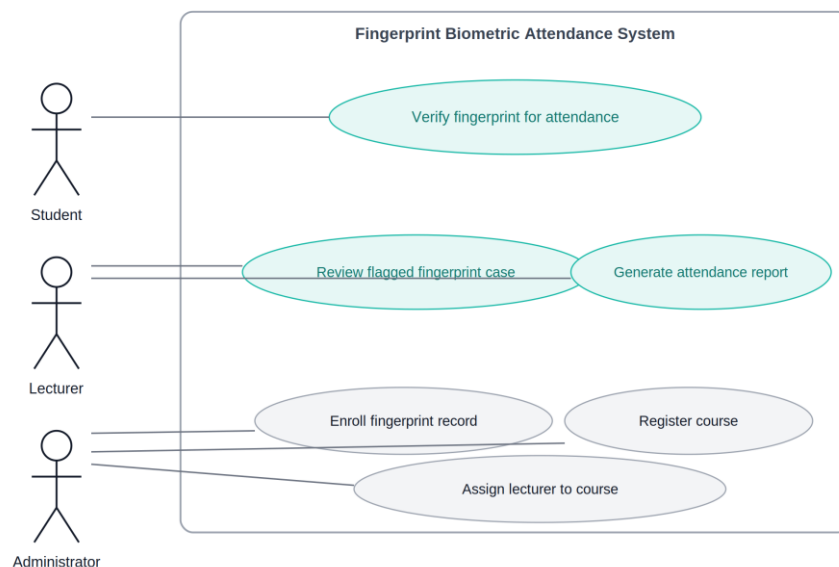


Figure 4.1: Use case diagram for the proposed fingerprint biometric attendance system

This diagram is a direct match to the functional requirements listed in Chapter Three, nothing has been added here that was not already stated there. Each oval in the diagram represents one thing the system must be able to do, and each line simply shows which actor is responsible for that action.

4.3.1 Use Case Descriptions

The table below sets out, for each use case shown in Figure 4.1, who is responsible for it and what happens when it is carried out. This is meant to make the diagram easier to follow, not to add anything beyond what it already shows.

Table 4.0: Use Case Descriptions

Use Case	Responsible Actor	What Happens
Verify fingerprint for attendance	Student	A student places a finger on the reader during a class session. The system checks the scan against stored templates and creates an attendance entry, marked as normal if a match is found or flagged if it is not.
Review flagged fingerprint case	Lecturer	A lecturer looks at any attendance entry marked as flagged for their course and decides how to handle it, for example by confirming the student was present through another means.
Generate attendance report	Lecturer	A lecturer requests a summary of attendance for their course, showing each student's attendance record over a given period.
Enroll fingerprint record	Administrator	An administrator enters a new student or staff member's basic details and captures their fingerprint, creating a new stored record.
Register course	Administrator	An administrator adds a new course to the system, including its title.
Assign lecturer to course	Administrator	An administrator links a staff member to a course as its lecturer.

4.4 Entity Relationship Diagram

The diagram below shows how the proposed system's data would be organised. Four main entities are involved. A User covers a student, a staff member, or an administrator, since all three are people the system needs to keep a record of, told apart by a role field. A Fingerprint Template stores the processed fingerprint data captured during enrollment for a given user, a Course stores basic course information along with the user acting as its lecturer, and an Attendance Record stores one entry each time a user is marked present for a course on a given date.

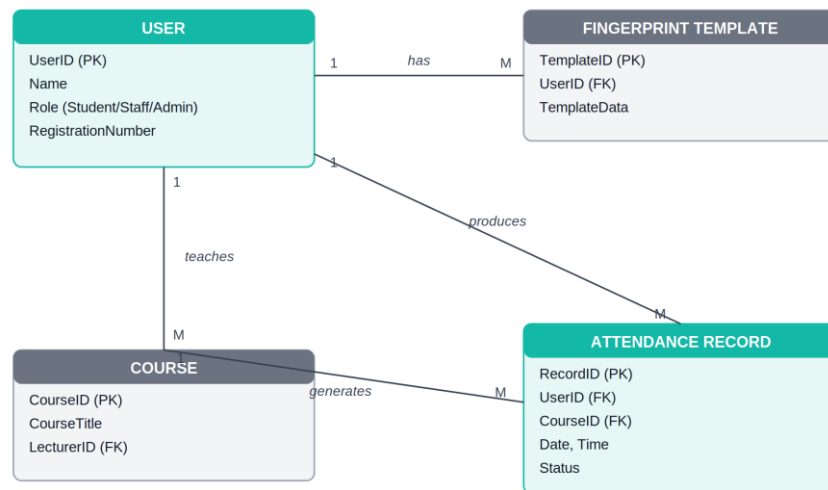


Figure 4.2: Entity relationship diagram for the proposed system

This study allows a user to have more than one stored fingerprint template as one way of addressing the enrollment-failure gap identified in Chapter Two, where none of the reviewed studies set out a specific plan for what to do when a person's fingerprint cannot be read reliably. A user can also teach more than one course, and a course can have many attendance records built up over a semester, one for each time a class is held. A user, whether student or staff, can likewise build up many attendance records over time. None of these relationships go beyond what a fingerprint attendance system of this kind actually needs, kept deliberately simple rather than padded out with entities the proposed system has no real use for.

The relationships between these tables are what make the design work rather than just a list of fields. UserID appears in the FingerprintTemplate table as a foreign key, meaning every stored fingerprint template is tied directly to one specific user record. This is what allows one user to have more than one template enrolled, which is the design's answer to the failure-to-enroll gap discussed in Chapter Two. A student whose dominant finger cannot be reliably read can enroll a second finger, and both templates point back to the same UserID without creating a duplicate user record. UserID and CourseID both appear in the AttendanceRecord table for the same reason: every attendance entry must be linked to a specific person and a specific course. An entry cannot exist without both, which prevents the kind of ambiguous or incomplete records that the manual process regularly produces.

4.5 Database Design

Based on the entity relationship diagram above, the proposed database would need four tables. The fields listed below are the ones needed to support the functional requirements from Chapter Three, and no more.

Table 4.1: User

Field	Data Type	Description
UserID	Integer	A unique number identifying each user, used as the primary key.
Name	Text	The full name of the student, staff member, or administrator.
Role	Text	Identifies the user as a student, staff member, or administrator.
RegistrationNumber	Text	The student's matriculation number or the staff member's employee number.

Table 4.2: FingerprintTemplate

Field	Data Type	Description
TemplateID	Integer	A unique number identifying each stored fingerprint

		template.
UserID	Integer	Links the template to the user it belongs to.
TemplateData	Binary data	The processed fingerprint data produced during enrollment, used for matching during verification.

Table 4.3: Course

Field	Data Type	Description
CourseID	Integer	A unique number identifying each course.
CourseTitle	Text	The name of the course.
LecturerID	Integer	Links the course to the staff member assigned to teach it.

Table 4.4: AttendanceRecord

Field	Data Type	Description
RecordID	Integer	A unique number identifying each attendance entry.
UserID	Integer	Links the entry to the student or staff member marked present.
CourseID	Integer	Links the entry to the course the attendance was taken for.
Date	Date	The date the attendance was recorded.
Time	Time	The time the attendance was recorded.
Status	Text	Shows whether the entry is a normal match or a flagged, unmatched case.

4.6 System Process Flow

4.6.1 Enrollment Process

Enrollment would happen once for each student or staff member, carried out by an administrator. First, the administrator enters the person's basic details, name, role, and registration number, into the system. Next, the person places their finger on the fingerprint reader, and the system captures the image and processes it into a template. To help address the enrollment-failure gap identified in Chapter Two, a second finger could also be enrolled for the same person, so that an alternative is available if the first fingerprint later becomes difficult to read. Once this is done, the new record is stored and the person is ready to use the system to mark attendance.

4.6.2 Attendance Verification Process

Verification would happen every time a class is held. A lecturer starts an attendance session for their course, and each student places their finger on the reader in turn. The system captures the new scan and compares it against the templates already stored for that course's registered students. If a match is found, the system creates a new attendance record with the current date, time, and course, and marks the entry as normal. If no match is found, rather than simply marking the person absent, the system creates the entry with a flagged status instead, so that a staff member can look into the case afterward, whether it turns out to be a genuine problem with the person's fingerprint or an attempt to use the system incorrectly. This flagging step is the direct answer to the gap identified in Chapter Two, where most of the reviewed studies did not explain what their systems do when a fingerprint cannot be matched.

The diagram below sets out this same process as a flow, showing the point at which the system decides between a normal entry and a flagged one.

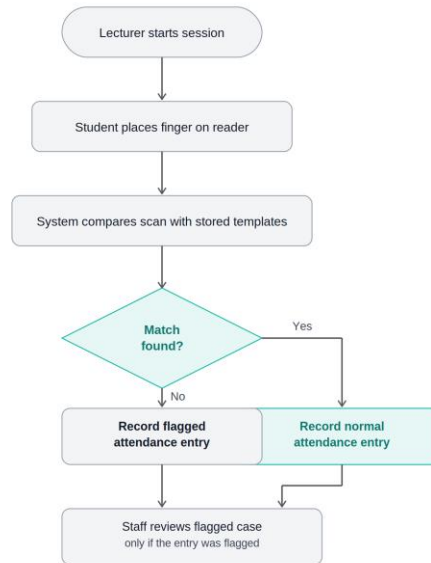


Figure 4.3: Attendance verification process flow

4.7 Input and Output Design

This section describes, in general terms, what the main screens of the proposed system would show. No specific software has been chosen, so this is a description of the information each screen would need to hold, not a fixed layout.

The enrollment screen, used by an administrator, would need fields for the person's name, their role, their registration number, and a button to begin capturing a fingerprint through the connected reader. Once a fingerprint scan is accepted, the screen would confirm that the record has been saved.

The attendance report, generated for a lecturer or an administrator, would need to show the course name, the date range covered, and a list of students with their attendance status for each session in that period, normal, absent, or flagged. A summary line showing each student's overall attendance percentage against the institution's minimum requirement would sit alongside this list, since this is the figure examination eligibility is ultimately judged against.

The percentage shown in that summary line would be calculated by dividing the number of sessions a student was marked present by the total number of sessions held for that course in the period covered by the report, then multiplying by one hundred. A student marked present in

fourteen out of twenty sessions, for example, would show 70 percent, which falls below the National Universities Commission's 75 percent threshold and would appear as a student at risk of examination disqualification. A student marked present in sixteen out of twenty sessions would show 80 percent and would be clear. The report does not make the eligibility decision itself, that remains with academic staff and the registry, but it gives them an accurate, automatically calculated percentage to base that decision on, which is something the manual process, relying on end-of-semester handwritten tallies, cannot reliably produce.

4.8 Summary

This chapter has presented the design of the proposed fingerprint biometric attendance system, covering its overall architecture, a use case diagram and description showing what each type of user can do, an entity relationship diagram showing how its data would be organized, a simple database design based on that diagram, a description and flowchart of how enrollment and attendance verification would work in practice, and an outline of the system's main input and output screens. Every part of this design traces back to a requirement already set out in Chapter Three, and no feature has been added beyond what those requirements call for. Taken together, this chapter is this study's complete answer to its third research question. The next chapter summarizes this study and sets out its conclusions and recommendations.

CHAPTER FIVE

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

5.1 Introduction

This chapter brings the study to a close. It summarizes what each chapter has covered, draws a conclusion from the work carried out, states what this study has contributed, sets out recommendations, and points to areas a future study could take further. Nothing in this chapter introduces new material, it draws together what has already been presented in Chapters One to Four.

5.2 Summary of the Study

This study set out to design a fingerprint biometric attendance system for Mountain Top University, an institution the researcher did not have direct access to at the time of the study. Chapter One set out the background to this problem, explaining that manual attendance recording in Nigerian universities is tied directly to examination eligibility through the National Universities Commission's minimum attendance rule, and that this manual process is widely documented as slow, easy to falsify, and heavy on administrative staff. Chapter One also stated the aim and objectives of the study, its research questions, its scope, and its limitations, the most important of these being that this study relies entirely on secondary data rather than data collected directly from Mountain Top University.

Chapter Two reviewed twenty-four studies on fingerprint and other automated attendance systems, checked one at a time to confirm they were real, distinct, and relevant. Fifteen of these were carried out at Nigerian universities and polytechnics, six at institutions outside Nigeria, and three took a different approach: one using only RFID cards, one using facial recognition, and one reviewing hardware across the literature rather than building a new system. This review found that the problems these studies set out to solve, impersonation, wasted class time, and unreliable manual records, are consistent across more than a decade of research and across many different institutions. It also found real gaps in this literature, most notably that very few of the studies explain what their system does when a fingerprint cannot be matched, a gap this study treated as something to design around rather than ignore.

Chapter Three explained the methodology this study follows, a design and development research approach built on secondary data, and set out the functional and non-functional requirements the proposed system would need to meet. It also explained why Object-Oriented Analysis and Design was chosen over alternatives such as Waterfall and Agile, given that this study stops at the design stage and does not build or test a working system.

Chapter Four presented the actual design, an overall system architecture made up of an enrollment part, a verification part, a database, and a reporting part, a use case diagram and description showing what a student, a lecturer, and an administrator can each do, an entity relationship diagram and a simple database design showing how the system's data would be organised, and a description and flowchart of how enrollment and attendance verification would work in practice.

5.3 Conclusion

Based on the literature reviewed and the design work carried out, this study concludes that a fingerprint biometric attendance system is a reasonable and well-supported response to the weaknesses of manual attendance recording. The literature reviewed in Chapter Two shows, across many different institutions and more than a decade of research, that fingerprint systems consistently reduce the kind of impersonation and time wastage associated with paper-based methods. Nothing in this study's own literature review or design work contradicts that pattern, and the design set out in Chapter Four was built specifically to carry that pattern forward for Mountain Top University while also addressing a gap the existing literature mostly leaves unanswered, what a system should do when a fingerprint cannot be matched.

This conclusion comes with an honest limit already stated in Chapter One and worth repeating plainly here. This study has not built or tested the design it proposes, and it cannot report on how the system would actually perform once implemented, since no implementation was carried out. The conclusion above is a conclusion about the strength of the case for such a system, drawn from existing evidence, not a claim that the specific design in Chapter Four has itself been proven to work.

5.4 Contribution to Knowledge

This study contributes to the existing body of research on fingerprint biometric attendance systems in three ways. First, it proposes a design specifically for Mountain Top University, an institution not previously covered by this literature, whereas the twenty-four studies reviewed in Chapter Two were each carried out at institutions with their own direct researcher access. Second, it approaches this problem entirely through secondary literature, checked and verified one source at a time, offering a model for how a design-stage study can be carried out honestly when direct access to an institution is not available. Third, by identifying and responding to the enrollment-failure gap found across most of the reviewed literature, this study's proposed design includes a specific answer, flagging an unmatched fingerprint for staff review rather than silently marking a person absent, to a question most of the existing literature leaves open.

5.5 Recommendations

Based on the work carried out in this study, the following recommendations are made:

- i. Mountain Top University, or a similar institution considering a move away from manual attendance recording, should treat the design set out in Chapter Four as a starting point for a future implementation project, one with direct access to the institution's staff, students, and existing infrastructure.
- ii. any future implementation should budget time and resources for handling enrollment failure specifically, rather than assuming every fingerprint will be read successfully on the first attempt, since the literature reviewed in Chapter Two shows this is a real, if uncommon, occurrence.
- iii. before full deployment, a future implementation should be tested on a small scale first, in one department or one college, so that problems specific to Mountain Top University's own lecture halls, class sizes, and network conditions can be identified before the system is rolled out more widely.
- iv. the institution should put in place a clear policy on how long fingerprint data is stored and who may access it, in keeping with the kind of data protection concerns that come with holding biometric information on students and staff.

v. before committing to any implementation, the institution should assess its own infrastructure honestly, including the reliability of its power supply, the quality of its network connection in lecture halls, and its capacity to maintain the fingerprint sensors regularly. The literature reviewed in Chapter Two shows that a fingerprint attendance system is technically feasible for a Nigerian university setting, but also that power outages, network drops, and sensor degradation from repeated use in warm and humid conditions are real constraints that do not appear in most published studies. Planning for these from the start is more practical than treating them as problems to solve after deployment.

5.6 Suggestions for Further Studies

This study is limited to a design proposed from secondary literature, and several natural next steps follow from that limit. A future study could take the design presented in Chapter Four and actually implement it, testing it with real students and staff at Mountain Top University to see how closely its performance matches what has been reported in the literature reviewed here. Such a study would also be in a position to test the enrollment-failure handling proposed in this design directly, something this study could only argue for rather than demonstrate.

A further study could also investigate, specifically at Mountain Top University, the relationship between attendance and academic performance, following the direction taken by Adenekan and colleagues (2026) at Moshood Abiola Polytechnic, to see whether more accurate attendance recording through a fingerprint system produces any measurable change in student outcomes at this particular institution. Either of these directions would move this area of research from the design stage this study has completed toward the kind of tested, implemented system that most of the literature reviewed in Chapter Two already represents for other institutions.

A third direction concerns hardware standardization. The critical analysis in Chapter Two found that the studies reviewed use several different fingerprint sensor modules and microcontrollers without explaining the effect those choices have on the accuracy figures they report. A future study that tests two or more sensor types under the same conditions at a Nigerian university, measuring how false rejection rate changes between sensors in a real lecture hall environment,

would fill a gap that currently makes it difficult to compare results across the existing literature with any confidence.

5.7 Summary of the Chapter

This chapter has summarized the study from Chapter One through Chapter Four, drawn a conclusion that a fingerprint biometric attendance system is a well-supported response to the weaknesses of manual attendance recording while being honest about the limits of a design-only study, stated this study's contribution to existing knowledge, and set out recommendations and suggestions for further research. This brings the study to its close.

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